

Minkowski Geometry

Special relativity is Euclidean geometry with one sign flipped. Everything strange about it is that sign, working.

WHERE WE ARE

Chapter 2.2 produced a transformation and then a pile of consequences: simultaneity fails, moving clocks run slow, moving rods are short, velocities do not add, rapidities do. Each was derived, each is correct, and taken together they are a list. Lists are how a subject looks before anyone has found its organising idea.

This chapter finds the organising idea. It is already sitting in 2.2 §2.4, noticed in passing and then left alone: every one of those transformations preserves the combination

$$\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

One number, agreed on by everybody. We are going to take that seriously and see what it forces. What it forces is a **geometry** — a four-dimensional space with a notion of distance, in which the Lorentz transformations are the rigid motions, boosts are rotations, worldlines have lengths, and the length of a worldline is the time a clock carried along it reads. After that, the list stops being a list. Time dilation is a statement about lengths. The twin paradox is a statement about triangles. The light barrier is a statement about which directions exist.

Here is the thesis in one sentence, and it is worth reading twice: **special relativity is Euclidean geometry with one sign flipped**. Nearly every difference between the two subjects — why boosts are hyperbolic rather than circular, why the straight path is the *longest* and not the shortest, why some pairs of distinct points are at zero distance, why there is a speed limit at all — traces back to a single minus sign in the distance formula. We will flip it, and then follow it everywhere it goes.

One thing this chapter does *not* do honestly: it writes x^μ , $\eta_{\mu\nu}$ and $\Delta s^2 = \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu$, and uses index notation informally, without defining what an index means. That debt is real, it is flagged where it is incurred, and Chapter 2.4 pays it in full.

TOOLS YOU'LL NEED — Chapter 2.2: the boost $ct' = \gamma(ct - \beta x)$, $x' = \gamma(x - \beta ct)$, the matrix form, and the rapidity ϕ with $\beta = \tanh \phi$. Nothing from 2.2 is re-derived here. Chapter 0.4 §3 and §5: linear maps as matrices, composition as matrix multiplication, and the determinant — §1 asks which linear maps preserve a given quadratic expression, which is exactly a 0.4 question. Chapter 0.5 §1: the inner-product axioms, and the observation that once you have a bilinear form you have lengths, angles and orthogonality — this chapter removes one of those axioms on purpose and watches which conclusions survive. Chapter 1.2 §9, worked example 2: extremising $\int ds$ gives straight lines in the plane and great circles on a sphere. Section 6 runs that machine one more time, in spacetime.

1 • The invariant interval

Two events. Not two objects, not two particles — two *events*, each a point of spacetime with coordinates (t, x, y, z) in some inertial frame. Define

$$\Delta s^2 \equiv c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2, \quad (2.3.1)$$

where $\Delta t = t_2 - t_1$ and so on. This is called the **interval** between the two events, or sometimes the *spacetime separation*. Three warnings, all of which will be paid off before the chapter ends. The notation Δs^2 is a single symbol, not the square of something called Δs ; it can be negative. It is not a distance. And the minus signs are not a typographical accident — they are the entire subject.

1.1 • It is the same in every frame

Take the boost of Chapter 2.2,

$$ct' = \gamma(ct - \beta x), \quad x' = \gamma(x - \beta ct), \quad y' = y, \quad z' = z, \quad (2.3.2)$$

and apply it to a *separation* rather than to a single event. That is legitimate for exactly one reason, which 2.2 §1.2 worked hard for: the transformation is linear, so it acts on differences the same way it acts on coordinates. Now substitute and grind:

$$\begin{aligned} c^2 \Delta t'^2 - \Delta x'^2 &= \gamma^2 (c\Delta t - \beta\Delta x)^2 - \gamma^2 (\Delta x - \beta c\Delta t)^2 \\ &= \gamma^2 [c^2 \Delta t^2 - 2\beta c\Delta t \Delta x + \beta^2 \Delta x^2] \\ &\quad - \gamma^2 [\Delta x^2 - 2\beta c\Delta t \Delta x + \beta^2 c^2 \Delta t^2] \\ &= \gamma^2 (1 - \beta^2) c^2 \Delta t^2 - \gamma^2 (1 - \beta^2) \Delta x^2 \\ &= c^2 \Delta t^2 - \Delta x^2, \end{aligned} \quad (2.3.3)$$

using $\gamma^2(1 - \beta^2) = 1$, which is the definition of γ rearranged. The cross terms cancelled identically — they had to, because they enter the two brackets with the same coefficient and opposite overall sign. Since $\Delta y' = \Delta y$ and $\Delta z' = \Delta z$, we may subtract those too without changing anything, and

$$\boxed{c^2 \Delta t'^2 - \Delta x'^2 - \Delta y'^2 - \Delta z'^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.} \quad (2.3.4)$$

Every inertial observer, handed the same two events, computes the same Δs^2 . They will disagree about Δt . They will disagree about Δx . They will disagree about which event came first, sometimes. They agree about this.

Chapter 2.2 got here too, in its §2.4, and treated it as a consistency check — we demanded that light go at c , and the check confirmed that a spherical light pulse stays spherical. That reading undersells it badly. Postulate 2 constrains only the events with $\Delta s^2 = 0$, a single three-dimensional family out of all pairs. What came back out is a statement

about every pair of events, at every value of Δs^2 . The theory handed us more invariance than we asked for, and quantities that are invariant for no reason you demanded are exactly the quantities worth building a subject on.

1.2 · The reframe: this is what "length" means

Now the move that makes the rest of the book easier. Ask what a rotation is.

You probably answer "a rigid turn about an axis", which is a picture, not a definition. The definition used in mathematics — and the one Chapter 0.4 §5 was quietly preparing you for — is algebraic:

$$\text{a \textbf{rotation} of } \mathbb{R}^3 \text{ is a linear map that preserves } x^2 + y^2 + z^2. \quad (2.3.5)$$

That is the whole content. You do not need to mention turning, axes, or angles. Demand that a linear map leave $x^2 + y^2 + z^2$ unchanged for every vector, and rotations (together with reflections) are what you get; they form the group $O(3)$, and requiring the determinant to be $+1$ throws away the reflections. The logic runs *from* the quadratic form to the transformations, not the other way round. And the quantity that the transformations preserve is what we call **length** — that is not a separate fact about the world, it is the definition of length, namely the thing on which all observers with differently-oriented axes agree.

Read (2.3.4) again with that in mind. It says:

THE WHOLE CHAPTER, IN ONE SUBSTITUTION

Lorentz transformations are the linear maps that preserve Δs^2 .

So Δs^2 plays the role that squared length plays in Euclidean geometry, and the Lorentz transformations play the role that rotations play. Change the quadratic form from $x^2 + y^2 + z^2$ to $c^2 t^2 - x^2 - y^2 - z^2$, and the group of "rigid motions" changes from rotations to boosts-and-rotations. Nothing else about the logic changes at all.

That is not an analogy. It is the same definition with a different quadratic form in it.

Two remarks on how much this claims. First, the statement "Lorentz transformations preserve Δs^2 " is what we just proved; the statement "the linear maps preserving Δs^2 are exactly the Lorentz transformations" is a converse, and converses need proof. The grind box below proves it in the case that matters — one space dimension — and finds three extra maps hiding in the answer, which turn out to be parity and time reversal. Second, calling this a "geometry" commits us to lengths, angles, straight lines, and triangles behaving like a geometry. Sections 3 to 6 make good on each of those, and §6 finds that one of them behaves like a geometry *with the inequality reversed*. That is the sign flip, arriving where it hurts.

▼ Grind box — the converse: every linear map preserving $c^2 t^2 - x^2$, classified

Work in the (ct, x) plane, which is where all the content lives. Let

$$M = \begin{pmatrix} a & b \\ d & e \end{pmatrix}, \quad \begin{pmatrix} ct' \\ x' \end{pmatrix} = M \begin{pmatrix} ct \\ x \end{pmatrix},$$

and demand $c^2 t'^2 - x'^2 = c^2 t^2 - x^2$ for all (ct, x) . Expand the left side:

$$(act + bx)^2 - (dct + ex)^2 = (a^2 - d^2)c^2 t^2 + 2(ab - de)ctx + (b^2 - e^2)x^2.$$

Matching coefficients of the three independent monomials gives three equations:

$$a^2 - d^2 = 1, \quad ab - de = 0, \quad b^2 - e^2 = -1.$$

Solve the first. $a^2 = 1 + d^2 \geq 1$, so $|a| \geq 1$ and we may write $a = \sigma \cosh \phi$ with $\sigma = \pm 1$ and $\phi \geq 0$; then $d^2 = \sinh^2 \phi$, and by allowing ϕ to take either sign we can write $d = -\sigma \sinh \phi$ without loss of generality.

Solve the third. $e^2 = 1 + b^2 \geq 1$ likewise, so $e = \tau \cosh \psi$ and $b = -\tau \sinh \psi$ with $\tau = \pm 1$.

Use the second. Substituting,

$$ab - de = -\sigma\tau \cosh \phi \sinh \psi + \sigma\tau \sinh \phi \cosh \psi = \sigma\tau \sinh(\phi - \psi) = 0,$$

so $\psi = \phi$, since $\sinh$ vanishes only at zero. Therefore

$$M = \begin{pmatrix} \sigma \cosh \phi & -\tau \sinh \phi \\ -\sigma \sinh \phi & \tau \cosh \phi \end{pmatrix} = \begin{pmatrix} \sigma & 0 \\ 0 & \tau \end{pmatrix} \begin{pmatrix} \cosh \phi & -\sinh \phi \\ -\sinh \phi & \cosh \phi \end{pmatrix}.$$

The right-hand factor is exactly Chapter 2.2's boost in rapidity form. The left-hand factor is one of four sign choices: $\sigma = \tau = +1$ is the identity, $\sigma = +1, \tau = -1$ is **parity** $x \rightarrow -x$, $\sigma = -1, \tau = +1$ is **time reversal** $t \rightarrow -t$, and $\sigma = \tau = -1$ is both. Note $\det M = \sigma\tau$.

The conclusion. The linear maps preserving $c^2 t^2 - x^2$ are precisely the boosts, possibly composed with parity and/or time reversal. So the converse holds, up to those discrete factors — and this is the same structure as in Euclidean geometry, where preserving $x^2 + y^2$ gives rotations possibly composed with a reflection. The group falls into four disconnected pieces, and the piece containing the identity — $\sigma = \tau = +1$, the pure boosts — is the one physics uses without comment. It has a name, the *proper orthochronous* Lorentz group: proper means $\det = +1$, orthochronous means it does not reverse the direction of time. ■ That the same four-component structure survives in the full four-dimensional group, with rotations included, is quoted here and unpacked in Chapter 6.1.

A detail worth noticing. Nothing in this derivation used a postulate, a light ray, or a physical assumption. Given the quadratic form, the transformations are pure algebra. All the physics went into choosing $c^2 t^2 - x^2$ rather than $c^2 t^2 + x^2$, and Chapter 2.2 is where that choice was forced.

1.3 · The sign convention, and the one half the world uses instead

Nothing above depended on whether we write $c^2\Delta t^2 - \Delta x^2$ or $\Delta x^2 - c^2\Delta t^2$. The two differ by an overall factor of -1 , they are preserved by exactly the same transformations, and every physical statement can be phrased in either. Books nevertheless choose, and then never mention it again, which is how readers get ambushed.

This book uses the **timelike-positive** convention, often written $(+, -, -, -)$ and sometimes called the "mostly minus" or "particle physics" signature:

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2, \quad \text{so a clock's own path has } \Delta s^2 > 0. \quad (2.3.6)$$

The reason is Parts V to VII. In particle physics the object you write down forty times a day is the four-momentum, and its invariant square in this convention is $p \cdot p = m^2c^2$ — mass squared, positive, with no sign to remember. In the other convention it is $-m^2c^2$, and every mass-shell condition in the book would carry a minus sign for no benefit. Since Parts V–VII are much longer than Part III, we optimise for them.

The other convention, $(-, +, +, +)$ or "mostly plus", is standard in general relativity, where the object you write forty times a day is a spatial line element and you would rather *it* came out positive. If you open a relativity textbook and find $ds^2 = -c^2dt^2 + dx^2$, nothing is wrong; multiply by -1 and continue. What changes: the sign of Δs^2 classifying timelike versus spacelike, the sign in $p \cdot p$, and the sign of $\eta_{\mu\nu}$ itself. What does not change: a single physical prediction. It is a units-style choice, like measuring angles in degrees, and it carries the same amount of physics.

2 · Coordinates, the metric, and one line of index notation

We are about to need a compact way to write (2.3.1) that does not degenerate into four-fold bookkeeping every time. The notation that does this is one of the great labour-saving devices in physics, and it is worth introducing now — informally, honestly labelled as informal — because §3 onward reads much better in it.

2.1 · One symbol for four coordinates

Group the four coordinates of an event into a single object with an index:

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct, x, y, z). \quad (2.3.7)$$

Greek indices μ, ν, ρ, σ run over 0, 1, 2, 3; Latin indices i, j, k over the spatial 1, 2, 3 only. The time coordinate is $x^0 = ct$ and not t , so that all four entries have the dimensions of length. That is not cosmetic: Chapter 2.2 §2.2 already showed that ct and x enter the boost *symmetrically*, and writing t instead would hide the symmetry behind a units conversion.

The superscript is an index, not a power: x^2 is the y -coordinate; if we mean the square of something we will write it so the context is unmissable. This collision of notation is genuinely annoying and universal, and everyone lives with it.

2.2 · The metric, and the summation convention

Now define a 4×4 array of numbers

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad (2.3.8)$$

called the **Minkowski metric**. It carries the minus signs of (2.3.1) and does nothing else. With it, the interval is

$$\Delta s^2 = \sum_{\mu=0}^3 \sum_{\nu=0}^3 \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu \equiv \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu. \quad (2.3.9)$$

The second form uses the **Einstein summation convention**: *an index appearing twice in a single term, once up and once down, is summed over its whole range, and the summation sign is not written*. It is a convention about typography and nothing more, but the saving is substantial — the double sum has sixteen terms, of which twelve vanish because η is diagonal, and you never want to write that out twice.

Check that (2.3.9) says what it should. The surviving terms are $\mu = \nu$:

$$\eta_{\mu\nu} \Delta x^\mu \Delta x^\nu = (+1)(\Delta x^0)^2 + (-1)(\Delta x^1)^2 + (-1)(\Delta x^2)^2 + (-1)(\Delta x^3)^2, \quad (2.3.10)$$

which is $c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 \checkmark$. The metric is doing exactly one job: supplying a sign for each term. In ordinary three-dimensional space the corresponding array is $\delta_{ij} = \text{diag}(1, 1, 1)$, which supplies all plus signs, which is why nobody ever writes it and why the metric is invisible in first-year physics. **The whole of the difference between Euclidean geometry and Minkowski geometry is the difference between δ_{ij} and $\eta_{\mu\nu}$.**

▼ Grind box — the summation convention, written out once in full, and the rules for using it

The unabbreviated version of (2.3.9), all sixteen terms, with $\Delta x^\mu = (a, b, c, d)$ to avoid symbol collisions:

$$\begin{aligned} \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu &= \eta_{00}aa + \eta_{01}ab + \eta_{02}ac + \eta_{03}ad \\ &\quad + \eta_{10}ba + \eta_{11}bb + \eta_{12}bc + \eta_{13}bd \\ &\quad + \eta_{20}ca + \eta_{21}cb + \eta_{22}cc + \eta_{23}cd \\ &\quad + \eta_{30}da + \eta_{31}db + \eta_{32}dc + \eta_{33}dd \\ &= a^2 - b^2 - c^2 - d^2. \end{aligned}$$

Three rules, all of which will be justified properly in Chapter 2.4 and all of which you should obey immediately.

(1) A repeated index is dead. In $\eta_{\mu\nu}\Delta x^\mu\Delta x^\nu$ the labels μ and ν have been summed away; the result carries no index and does not depend on what you called them. Renaming $\mu \rightarrow \alpha$ throughout changes nothing. Such indices are called *dummy* indices, and they behave exactly like the k in $\sum_k a_k$.

(2) An unpeated index is alive, and must match on both sides. $V^\mu = \Lambda^\mu{}_\nu W^\nu$ is a legal equation: ν is summed, μ is free and appears once on each side, so the equation is four equations. $V^\mu = W^\nu$ is not an equation at all — it does not say which component equals which.

(3) Never use the same letter three times. $\eta_{\mu\mu}\Delta x^\mu$ is ill-formed: the convention cannot tell which pair you meant to sum. If you need two separate sums, use two separate letters. This rule catches more algebra errors than any other.

What is *not* being claimed here. We have written some indices up (x^μ) and some down ($\eta_{\mu\nu}$), and given no reason. There is a reason, it is important, and it is not that one is a row and the other a column. Chapter 2.4 §3 gives it. Until then, treat the placement as a spelling rule you are obeying on trust, and note that the trust is being tracked: every expression in this chapter has each summed index appearing once up and once down, and you may check that it does.

2.3 · What a four-vector is, provisionally

The displacement between two nearby events, $dx^\mu = (c dt, dx, dy, dz)$, transforms under a boost in a completely definite way — it is what (2.3.2) does. We take that as a template:

WORKING DEFINITION — FOUR-VECTOR

A **four-vector** is a set of four quantities V^μ , one per frame, that transform between frames *the same way* dx^μ does:

$$V'^\mu = \Lambda^\mu{}_\nu V^\nu,$$

with $\Lambda^\mu{}_\nu$ the same matrix that transforms coordinate displacements.

This is deliberately informal. It says "these ones, like that one", which is a definition by example, and it leaves at least three questions unanswered: why some indices are up and others down, what the object V^μ is as opposed to what its components do, and whether the definition depends on the transformation being a Lorentz transformation. Chapter 2.4 answers all three and turns this paragraph into a theorem. Nothing in this chapter needs the answers; everything in this chapter needs the template.

Why bother now? Because the payoff arrives immediately and repeatedly. If V^μ and W^μ are both four-vectors, then the combination

$$V \cdot W \equiv \eta_{\mu\nu} V^\mu W^\nu = V^0 W^0 - V^1 W^1 - V^2 W^2 - V^3 W^3 \quad (2.3.11)$$

is a single number that every observer agrees on — for exactly the reason Δs^2 is, since Δs^2 is the special case $V = W = \Delta x$. That is a machine for manufacturing invariants, and we will feed it: Δx^μ in this chapter, the four-velocity u^μ in §7, the four-momentum p^μ in Chapter 2.5 (where $p \cdot p = m^2 c^2$ turns out to be the whole of relativistic kinematics), the four-current in Chapter 2.6.

2.4 · The condition on Λ

One more line, and it is the line Chapter 2.4 and Chapter 6.1 both build on. Write the transformation in matrix form, as Chapter 2.2 §3 did, and demand invariance of the interval. In index notation:

$$\eta_{\mu\nu} \Delta x'^\mu \Delta x'^\nu = \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \Delta x^\rho \Delta x^\sigma \stackrel{!}{=} \eta_{\rho\sigma} \Delta x^\rho \Delta x^\sigma. \quad (2.3.12)$$

Since the separation Δx^ρ is arbitrary and both sides are symmetric in $\rho \leftrightarrow \sigma$, the coefficients must match term by term:

$$\boxed{\eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = \eta_{\rho\sigma}, \quad \text{that is} \quad \Lambda^\top \eta \Lambda = \eta.} \quad (2.3.13)$$

Stare at $\Lambda^\top \eta \Lambda = \eta$ and compare it with the condition defining a rotation matrix, $R^\top 1 R = 1$, i.e. $R^\top R = 1$. Same equation, one symbol different. ■ (2.3.13) is the *defining equation of the Lorentz group*, written $O(1, 3)$ — the notation records that the metric has one plus and three minuses — and Chapter 6.1 will take it as the starting point rather than the conclusion, differentiate it near the identity, and read off the six generators. Problem 4 does the counting by hand and gets six.

Two immediate consequences, both free. Taking determinants of (2.3.13) and using $\det(AB) = \det A \det B$ from Chapter 0.4 §5,

$$(\det \Lambda)^2 \det \eta = \det \eta \quad \implies \quad \det \Lambda = \pm 1, \quad (2.3.14)$$

so Lorentz transformations preserve four-dimensional volume in spacetime up to sign — Chapter 2.2 §3 checked $\det = +1$ for the boost explicitly. And the set of matrices satisfying (2.3.13) is closed under multiplication and inversion, which is the group property Chapter 2.2 §3.1 verified by brute force for boosts along one axis; here it is one line, because $(\Lambda_2 \Lambda_1)^\top \eta (\Lambda_2 \Lambda_1) = \Lambda_1^\top (\Lambda_2^\top \eta \Lambda_2) \Lambda_1 = \Lambda_1^\top \eta \Lambda_1 = \eta$.

▼ Grind box — checking $\Lambda^\top \eta \Lambda = \eta$ on the boost, entry by entry

Only the upper-left 2×2 block does anything, so drop y and z and take $\eta = \text{diag}(1, -1)$ with

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix} = \begin{pmatrix} \cosh \phi & -\sinh \phi \\ -\sinh \phi & \cosh \phi \end{pmatrix}.$$

The boost matrix is symmetric, so $\Lambda^\top = \Lambda$. Compute $\eta \Lambda$ first — multiplying on the left by $\text{diag}(1, -1)$ flips the sign of the second row:

$$\eta\Lambda = \begin{pmatrix} \cosh \phi & -\sinh \phi \\ \sinh \phi & -\cosh \phi \end{pmatrix}.$$

Then

$$\begin{aligned} \Lambda^\top \eta \Lambda &= \begin{pmatrix} \cosh \phi & -\sinh \phi \\ -\sinh \phi & \cosh \phi \end{pmatrix} \begin{pmatrix} \cosh \phi & -\sinh \phi \\ \sinh \phi & -\cosh \phi \end{pmatrix} \\ &= \begin{pmatrix} \cosh^2 \phi - \sinh^2 \phi & -\cosh \phi \sinh \phi + \sinh \phi \cosh \phi \\ -\sinh \phi \cosh \phi + \cosh \phi \sinh \phi & \sinh^2 \phi - \cosh^2 \phi \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \eta \quad \checkmark \end{aligned}$$

The diagonal entries are the hyperbolic Pythagorean identity; the off-diagonal ones cancel because the matrix is symmetric. Note which identity did the work: $\cosh^2 - \sinh^2 = 1$. In the Euclidean case the same computation with $\eta \rightarrow 1$ and $\Lambda \rightarrow R(\theta)$ runs on $\cos^2 + \sin^2 = 1$ instead. One sign, two geometries — which is §3.

3 · Boosts are hyperbolic rotations

This is the section the chapter exists for. Everything before it was setup and everything after it is consequence.

3.1 · Two matrices, side by side

Put the Euclidean rotation and the Lorentz boost next to each other and read across.

A rotation in the xy plane preserves $x^2 + y^2$ and is built from a pair of functions satisfying $\cos^2 \theta + \sin^2 \theta = 1$:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \quad x'^2 + y'^2 = x^2 + y^2. \quad (2.3.15)$$

A boost in the (ct, x) plane preserves $c^2 t^2 - x^2$ and is built — this is Chapter 2.2 §6.1 — from a pair of functions satisfying $\cosh^2 \phi - \sinh^2 \phi = 1$:

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \cosh \phi & -\sinh \phi \\ -\sinh \phi & \cosh \phi \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}, \quad c^2 t'^2 - x'^2 = c^2 t^2 - x^2. \quad (2.3.16)$$

The parallel is not loose. Each matrix is built from the two functions whose squares obey the identity that matches the quadratic form; each preserves that form; each is parametrised by one real number. And the parameter is additive in both cases:

$$R(\theta_2)R(\theta_1) = R(\theta_1 + \theta_2), \quad \Lambda(\phi_2)\Lambda(\phi_1) = \Lambda(\phi_1 + \phi_2), \quad (2.3.17)$$

the first by the circular addition formulas, the second by the hyperbolic ones — Chapter 2.2 §6.2 multiplied the matrices out. So:

THE RAPIDITY IS THE ANGLE

The rapidity ϕ of Chapter 2.2 is not merely "the variable in which velocities add". It is the **hyperbolic angle of the rotation**, and it adds for exactly the reason Euclidean angles add: composing two rotations of the same kind about the same axis adds their angles, because that is what a one-parameter group of rotations does.

Chapter 2.2 arrived at ϕ by noticing that $\beta_3 = (\beta_1 + \beta_2)/(1 + \beta_1\beta_2)$ is the $\tanh$ addition theorem and taking the hint. Here the hint is explained. Velocity is $\beta = \tanh \phi$, and $\tanh$ of an angle is a *slope*. Slopes never add; angles do. Three centuries of physics added the slopes.

Two differences are worth flagging before we exploit them, because they are the places where the geometries part company.

The signs in the off-diagonal entries. The rotation matrix has $+\sin$ above the diagonal and $-\sin$ below; the boost has $-\sinh$ in both places. So the boost matrix is *symmetric* and the rotation is not. That is not cosmetic: Chapter 2.2's Problem 3 used precisely this to prove that two boosts in different directions leave a rotation behind.

The range of the parameter. θ is periodic — rotate by 2π and you are back where you started, and the orbit of a point closes. ϕ is not periodic and runs over all of $\mathbb{R}$; boost forever and you never return. $\cos$ and $\sin$ are bounded, $\cosh$ and $\sinh$ are not. The compactness of the rotation group and the non-compactness of the boosts is the deepest structural difference between them, and it is responsible for a great deal in Part VI. It is also the reason $\gamma = \cosh \phi$ has no upper bound while a rotation never stretches anything.

3.2 · The orbits: circles become hyperbolae

Now ask the question that turns algebra into a picture. *Take one point and apply every transformation in the family. What curve does it sweep out?*

Euclidean. Start at (x_0, y_0) and rotate through every θ . Since $x^2 + y^2$ is preserved, every image point satisfies $x^2 + y^2 = r^2$ with $r^2 = x_0^2 + y_0^2$ fixed. The orbit is a **circle**. And every point of that circle is reached, since the angle covers the full range. So: *the orbits of the rotation group are the circles centred on the origin*, and a circle is precisely a locus of constant distance from the origin. This is so familiar that it takes an effort to notice it is a theorem.

Minkowski. Start at (ct_0, x_0) and boost through every ϕ . Since $c^2t^2 - x^2$ is preserved, every image satisfies

$$c^2t^2 - x^2 = k \quad \text{with} \quad k = c^2t_0^2 - x_0^2 \text{ fixed.} \quad (2.3.18)$$

That is the equation of a **hyperbola** — a rectangular hyperbola with asymptotes $ct = \pm x$. Explicitly, applying (2.3.16) to a point written in the form $(ct_0, x_0) = (\lambda \cosh \phi_0, \lambda \sinh \phi_0)$ for $k = \lambda^2 > 0$ gives

$$\begin{aligned} ct' &= \lambda (\cosh \phi_0 \cosh \phi - \sinh \phi_0 \sinh \phi) = \lambda \cosh(\phi_0 - \phi), \\ x' &= \lambda (\sinh \phi_0 \cosh \phi - \cosh \phi_0 \sinh \phi) = \lambda \sinh(\phi_0 - \phi), \end{aligned} \tag{2.3.19}$$

so the boost simply slides the hyperbolic angle: $\phi_0 \rightarrow \phi_0 - \phi$. Compare the Euclidean statement that a rotation slides the polar angle, $\theta_0 \rightarrow \theta_0 - \theta$. Identical structure, and the parametrisation ($\cosh, \sinh$) is doing exactly the job that ($\cos, \sin$) does.

THE INVARIANT HYPERBOLAE ARE THE CIRCLES OF THIS GEOMETRY

The curves $c^2 t^2 - x^2 = k$ are the loci of events at fixed interval from the origin. They are what "a circle of radius r " means when the distance function has a minus sign in it, and the four families they fall into are the whole of §4's causal structure, drawn:

- $k > 0$: two branches opening upward and downward, entirely inside the light cone. The upper branch is the set of events at proper time $\sqrt{k}/c$ into the future of the origin.
- $k < 0$: two branches opening left and right, entirely outside the light cone, at proper distance $\sqrt{-k}$ from the origin.
- $k = 0$: the hyperbola degenerates into its own asymptotes, the two lines $ct = \pm x$ — the light cone. This is the one case with no Euclidean counterpart, because $x^2 + y^2 = 0$ has only the single point $x = y = 0$ as a solution while $c^2 t^2 - x^2 = 0$ has a whole pair of lines.

A boost moves points *along* these curves and never across them. Which is to say: a boost can change an event's time coordinate by a factor of a thousand and its position coordinate likewise, and cannot change its interval from the origin by one part in 10^{15} . The figure below is built to make you watch exactly that happen.

One consequence deserves saying out loud because it kills a persistent confusion. The light cone $k = 0$ is an orbit too — a boost slides a null point along the ray $ct = x$, multiplying both coordinates by $e^{-\phi}$ (put $k = 0$ into (2.3.19), or just note $\cosh \phi - \sinh \phi = e^{-\phi}$). It can shrink the coordinates toward zero or blow them up without bound, and it can never move the point off the line. That is Postulate 2 in geometric dress: the light cone is not merely *a* surface that happens to be invariant, it is the degenerate member of the family of invariant hyperbolae, and no continuous boost can carry a point across it.

3.3 · Why the boosted axes look wrong, and why the ticks are wrong too

Chapter 2.2's figure showed the primed axes scissoring symmetrically toward the light line, and readers invariably ask two questions about it. First: *why are the ct' and x' axes not perpendicular, if S' is a perfectly good frame with perpendicular axes?* Second: *where along those axes is "one second", and why is it not where I would put it?* Both have the same answer, and we can now give it.

Perpendicularity. In Euclidean geometry, two directions are perpendicular when their dot product vanishes. In this geometry the dot product is (2.3.11), with the minus signs, so "perpendicular" — the right word is **orthogonal** — means $\eta_{\mu\nu} V^\mu W^\nu = 0$. The ct' axis is the set of events with $x' = 0$, so it points along $T^\mu = (\cosh \phi, \sinh \phi, 0, 0)$; the x' axis is the set with $t' = 0$, pointing along $X^\mu = (\sinh \phi, \cosh \phi, 0, 0)$. Their Minkowski dot product is

$$T \cdot X = \eta_{\mu\nu} T^\mu X^\nu = \cosh \phi \sinh \phi - \sinh \phi \cosh \phi = 0. \quad (2.3.20)$$

They *are* orthogonal — exactly, at every ϕ . They do not look it, because your eye is applying the Euclidean dot product of the page, in which their product is $\cosh \phi \sinh \phi + \sinh \phi \cosh \phi = \sinh 2\phi \neq 0$. The axes are perpendicular in the geometry and not on the paper, and the paper is wrong. Note the pleasing detail that the two axes are mirror images in the line $ct = x$: swapping $\cosh \leftrightarrow \sinh$ maps one to the other. That reflection is what "scissoring symmetrically about the light line" means, and it is forced by the fact that light itself is orthogonal to *itself* — a null vector $N = (1, 1, 0, 0)$ has $N \cdot N = 0$.

Calibration. Where is the event "one second of S' time, at S' 's origin"? By definition it is at $x' = 0, ct' = 1$, so its interval from the origin is $c^2 t'^2 - x'^2 = 1$. Intervals are invariant, so in the unprimed frame it is the point where the ct' axis meets the invariant hyperbola $c^2 t^2 - x^2 = 1$. From the parametrisation, that point is

$$(ct, x) = (\cosh \phi, \sinh \phi) = (\gamma, \gamma\beta). \quad (2.3.21)$$

So **the invariant hyperbolae calibrate the axes**: to mark off unit ticks along any observer's time axis, intersect that axis with the hyperbolae $c^2 t^2 - x^2 = 1, 4, 9, \dots$; to mark off unit ticks along the space axis, intersect it with $x^2 - c^2 t^2 = 1, 4, 9, \dots$. It works for every observer, because the hyperbolae belong to no observer.

How far off is the eye? The unit tick on the unprimed ct axis sits at Euclidean distance 1 from the origin on the page. The unit tick on the ct' axis sits at page-distance

$$\sqrt{\cosh^2 \phi + \sinh^2 \phi} = \sqrt{\cosh 2\phi} = \sqrt{\gamma^2 + \gamma^2 \beta^2} = \gamma \sqrt{1 + \beta^2}. \quad (2.3.22)$$

At $\beta = 0.6$ — where $\phi = \operatorname{artanh} 0.6 = \ln 2$ exactly, a small gift — this is $\sqrt{1.25^2 + 0.75^2} = \sqrt{2.125} = 1.4577$. The primed observer's second is drawn 46% longer than yours. It is not longer. The page is lying, uniformly and predictably, by the factor $\gamma \sqrt{1 + \beta^2}$:

β	0.2	0.4	0.5	0.6	0.8	0.9
ϕ	0.2027	0.4236	0.5493	0.6931	1.0986	1.4722
page stretch	1.041	1.175	1.291	1.458	2.134	3.087

Once you know that, length contraction and time dilation become visible on a diagram rather than mysterious. Chapter 2.2 §4.3 said two observers "slice the same four-dimensional object at different angles"; the slices are the lines parallel to each observer's x axis, and the calibration is why the same physical rod, sliced two ways, is assigned two different lengths without anything being squeezed.

⚠ WHY THIS ISN'T OBVIOUS — THE PAGE IS EUCLIDEAN AND THE GEOMETRY IS NOT

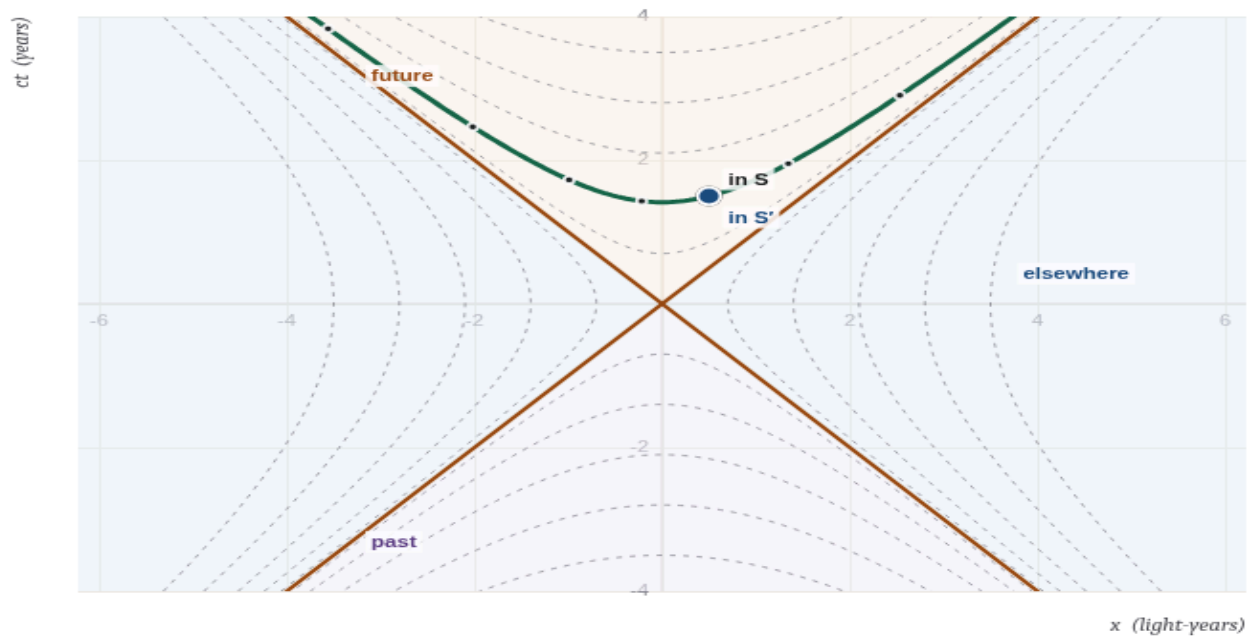
This is the single most common source of confusion when reading spacetime diagrams, so it is worth stating in its bluntest form: **on a spacetime diagram, longer on the page means shorter in proper time**. Your visual system measures distances with $\sqrt{\Delta x^2 + \Delta y^2}$ because that is the geometry of the paper. The diagram's content is $\sqrt{c^2 \Delta t^2 - \Delta x^2}$. These do not merely differ by a scale factor; they run in *opposite directions*.

Numbers make it concrete. Take Chapter 2.2's twins, in years and light-years, from $(ct, x) = (0, 0)$ to $(10, 0)$:

WORLDLINE	LENGTH ON THE PAGE	PROPER TIME
straight, at rest	10.00	10.00 yr
out to $x = 4$ at $\beta = 0.8$ and back	$2\sqrt{41} = 12.81$	6.00 yr
constant proper acceleration, same turning point	12.97	4.94 yr

Perfectly monotone, and backwards. The path that looks longest is the one along which least time elapses. Every drawn segment that tilts away from vertical is *buying* page-length with proper time, at a rate set by the minus sign. Train yourself to stop trusting the picture for lengths while continuing to trust it for everything else — incidence, ordering, which side of a light ray something is on — because those it gets right.

3.4 · Seeing it



rapidity ϕ

switch to Euclidean

reset angle

timelike event

spacelike event

null event

MINKOWSKI invariant $(ct)^2 - x^2$ rapidity $\phi = 0.0000$ $\beta = \tanh \phi = 0.00000$ $\gamma = \cosh \phi = 1.00000$

in S: $ct = +1.50000$ $x = +0.50000$ in S': $ct' = +1.50000$ $x' = +0.50000$

Δs^2 in S = +2.0000000000 | Δs^2 in S' = +2.0000000000 | difference = 0.00e+0

TIMELIKE, absolute future – proper time from the origin $\tau = 1.414214$ yr

One sign, two geometries. Inside this figure only, units are chosen with $c = 1$ — years for time, light-years for distance — so the vertical axis is ct and light travels at exactly 45° . Everything is computed numerically from $\cosh$, $\sinh$, $\cos$, $\sin$; no result is drawn from a formula stated elsewhere. **(i) Minkowski mode.** The faint curves are the invariant hyperbolae $c^2t^2 - x^2 = \pm k^2$; the orange lines are the light cone, the degenerate member $k = 0$. The hollow marker is an event as you placed it — **click or drag anywhere in the frame to move it** — and the solid marker is *the same event* as labelled by an observer boosted by rapidity ϕ . Slide ϕ and watch: the solid marker travels along the bold hyperbola through the event and never leaves it, while ct and x separately change by large factors. The readout gives Δs^2 computed independently in both frames; the two agree to the last digit shown. The small ticks along the bold curve are at rapidity intervals of 0.5, so you can see that the boost slides the point *uniformly in ϕ* — the hyperbola is the orbit, and ϕ is its natural parameter. **(ii) Euclidean mode.** The identical figure with one sign changed: the invariant is now $x^2 + y^2$, the level curves are circles, the transformation is an ordinary rotation, and the point slides along its circle instead. Ticks are at angle intervals of 0.5 rad. Flip between the two modes and you are watching the entire difference between Euclidean geometry and special relativity. **(iii)** Background shading in Minkowski mode marks the causal regions of §4 — absolute future, absolute past, and elsewhere — and the last readout line classifies the current event and reports its proper time from the origin (if timelike) or proper distance (if spacelike). Note that no amount of boosting ever moves the event out of the region it started in.

▼ Grind box — why ϕ deserves to be called an angle: it is twice a sector area, exactly as θ is

Calling ϕ an "angle" is so far only an analogy of algebraic form. Here is the fact that makes it more than that, and the computation is one line in each geometry.

Chapter 0.7 §5 derived Green's theorem, one corollary of which is that the area swept by the radius vector from the origin along a curve $(u(s), v(s))$ is

$$A = \frac{1}{2} \int (u \, dv - v \, du) = \frac{1}{2} \int (u\dot{v} - v\dot{u}) \, ds.$$

Circle. Parametrise the unit circle as $u = \cos s, v = \sin s$. Then

$$u\dot{v} - v\dot{u} = \cos^2 s + \sin^2 s = 1 \quad \implies \quad A = \frac{\theta}{2}.$$

Hyperbola. Parametrise the unit hyperbola $u^2 - v^2 = 1$ as $u = \cosh s, v = \sinh s$. Then

$$u\dot{v} - v\dot{u} = \cosh^2 s - \sinh^2 s = 1 \quad \implies \quad A = \frac{\phi}{2}.$$

The same integrand, the same value, the same conclusion: **the parameter is twice the area of the sector it sweeps out**, in both geometries. That is the honest sense in which ϕ is an angle. And it explains additivity without any trigonometry: applying two transformations sweeps two adjacent sectors, and *areas add*. Angles add because areas add — in the circular case and the hyperbolic case alike.

It also explains why the analogy could not have been avoided. "Angle" was never fundamentally about turning; it was about the invariant measure along the orbit of a one-parameter group, and both groups have one. Chapter 6.1 will call this measure the group parameter and will not care which geometry it came from.

A numerical check, since the claim is easy to test. Take $\phi = 1$. The sector bounded by the u -axis, the hyperbola, and the ray to $(\cosh 1, \sinh 1) = (1.5431, 1.1752)$ has area $\frac{1}{2}(1.5431)(1.1752) - \int_1^{1.5431} \sqrt{u^2 - 1} \, du = 0.9068 - 0.4068 = 0.5000$, which is $\phi/2$ ✓. (The integral evaluates to $\frac{1}{2}u\sqrt{u^2 - 1} - \frac{1}{2} \operatorname{arcosh} u$ between the limits.)

4 • Causal structure

In Euclidean space, $x^2 + y^2 + z^2$ is positive unless both points coincide. Every pair of distinct points is at a positive distance, and there is nothing more to say. Flip the sign and a new phenomenon appears immediately: Δs^2 can be positive, negative, or zero, and *which* it is turns out to be the most important thing about a pair of events.

4.1 · Three kinds of separation

THE CLASSIFICATION

For two events with interval $\Delta s^2 = c^2 \Delta t^2 - |\Delta \mathbf{x}|^2$:

- **Timelike** if $\Delta s^2 > 0$: $c |\Delta t| > |\Delta \mathbf{x}|$. Light has time to spare. A material particle can be present at both.
- **Null** (or *lightlike*) if $\Delta s^2 = 0$: $c |\Delta t| = |\Delta \mathbf{x}|$. Only light connects them, and only just.
- **Spacelike** if $\Delta s^2 < 0$: $c |\Delta t| < |\Delta \mathbf{x}|$. Not even light can get from one to the other.

The classification is **Lorentz-invariant**, and the proof is two words: Δs^2 is invariant, so the sign of Δs^2 is invariant. That is the entire argument, and its brevity is the point. Everything else in this section is unpacking what that one-line theorem means.

It is worth being clear about how strong it is. Whether two events are timelike separated is not a matter of who is looking, how fast they are moving, or which coordinates they chose. It is a fact about the pair, of exactly the same standing as "these two points are three metres apart" in Euclidean geometry — a frame-independent property, the first one we have met that is not just a number but a *structure*.

4.2 · The light cone

Fix an event O and ask which events are null-separated from it. With O at the origin, the condition $\Delta s^2 = 0$ reads

$$c^2 t^2 = x^2 + y^2 + z^2, \quad \text{that is} \quad |\mathbf{x}| = \pm ct. \quad (2.3.23)$$

At each time $t > 0$ this is a sphere of radius ct — the expanding flash of light emitted at O . Stack those spheres up the time axis and you get a cone; do the same for $t < 0$ and you get a second cone opening downward, the light that could have converged on O . Together they are the **light cone** of O . Suppress y and z and it is the pair of 45° lines in the figure of §3.4; suppress only z and it is an honest cone, which is where the name comes from.

The cone partitions spacetime into three regions, and every one of them is invariant, because the cone is invariant and the sign of Δt inside it is invariant (§4.3):

REGION	CONDITION	MEANING
Absolute future	$\Delta s^2 \geq 0$ and $\Delta t > 0$	Events O can influence. Every observer agrees they happen after O .
Absolute past	$\Delta s^2 \geq 0$ and $\Delta t < 0$	Events that can influence O . Every observer agrees they happen before O .
Elsewhere	$\Delta s^2 < 0$	No causal contact in either direction. Time order is frame-dependent.

The word *absolute* is doing real work and is not decoration: it marks the parts of "before" and "after" that survive the loss of absolute simultaneity. Chapter 2.2 destroyed the universal "now", and it is easy to conclude from that wreckage that time order is entirely conventional. It is not. Precisely the causally connectable part of it is absolute — which is the minimum required for physics to make sense, and, as we are about to see, exactly the maximum available.

"Elsewhere" is the genuinely new region, and it has no Newtonian counterpart. In Galilean physics every event is either before, after, or simultaneous with O , and the three cases exhaust everything. Here the middle case has swollen from a three-dimensional slice into a four-dimensional region, and it contains, at this instant, everything in the universe more than a few light-nanoseconds away from you. The Andromeda galaxy has no "now". It has a two-million-year-thick band of events about which your question "what is happening there right now?" has no frame-independent answer.

4.3 · The causality theorem

Claim. If two events are timelike or null separated, every inertial observer agrees on which one happened first.

Proof. Suppose $\Delta s^2 \geq 0$, so

$$c^2 \Delta t^2 \geq \Delta x^2 + \Delta y^2 + \Delta z^2 \geq \Delta x^2 \quad \implies \quad \left| \frac{\Delta x}{c \Delta t} \right| \leq 1, \quad (2.3.24)$$

where we may divide because $\Delta t \neq 0$ (if $\Delta t = 0$ then $\Delta s^2 \leq 0$ with equality only for coincident events). Choose axes so the boost is along x ; that costs nothing, since a rotation changes neither Δt nor the classification. Then, from (2.3.2), factor $c \Delta t$ out:

$$c \Delta t' = \gamma (c \Delta t - \beta \Delta x) = \gamma c \Delta t \left[1 - \beta \frac{\Delta x}{c \Delta t} \right]. \quad (2.3.25)$$

The bracket is bounded below:

$$1 - \beta \frac{\Delta x}{c \Delta t} \geq 1 - |\beta| \left| \frac{\Delta x}{c \Delta t} \right| \geq 1 - |\beta| > 0, \quad (2.3.26)$$

using $|\beta| < 1$, which is not an extra assumption but a property of every Lorentz transformation. Since $\gamma > 0$ and the bracket is strictly positive, $\Delta t'$ has the same sign as Δt in every frame. ■

Notice exactly where the proof lives: in the collision of two inequalities, $|\Delta x/c\Delta t| \leq 1$ from the separation being timelike, and $|\beta| < 1$ from the observer being an observer. Neither alone is enough. If observers could exceed c , or if influences could be spacelike, the bracket could reach zero and change sign. Causality is protected by the *product* of two things being less than one.

A sharper version, free of charge. Squaring the invariance relation, $c^2\Delta t'^2 = \Delta s^2 + |\Delta \mathbf{x}'|^2 \geq \Delta s^2$, so

$$c |\Delta t'| \geq \sqrt{\Delta s^2} \quad \text{for every frame,} \quad (2.3.27)$$

with equality exactly in the frame where $\Delta \mathbf{x}' = 0$. So for a timelike pair, $\Delta t'$ does not merely fail to vanish — it is bounded away from zero by a frame-independent amount, and no observer can even make the two events *nearly* simultaneous. Worked example 1 puts numbers on this.

4.4 · Spacelike: the order is genuinely up for grabs

Now the other case, and it is not a near miss. **Claim.** If two events are spacelike separated, there is a frame in which they are simultaneous, a frame in which A precedes B , and a frame in which B precedes A .

Proof. Take $\Delta s^2 < 0$ and rotate axes so the spatial separation lies along x ; then $\Delta y = \Delta z = 0$ and $c^2\Delta t^2 < \Delta x^2$, so $\Delta x \neq 0$ and

$$\left| \frac{c \Delta t}{\Delta x} \right| < 1. \quad (2.3.28)$$

From (2.3.25), $\Delta t'$ vanishes when

$$\beta_0 = \frac{c \Delta t}{\Delta x}, \quad (2.3.29)$$

and by (2.3.28) that value satisfies $|\beta_0| < 1$: it is an allowed velocity, so the frame exists. Since $c\Delta t' = \gamma(c\Delta t - \beta\Delta x)$ is a continuous, strictly monotonic function of β that passes through zero at β_0 (its derivative in β never vanishes for spacelike separations), it is positive on one side and negative on the other. Boosts with β slightly less than β_0 give one order; slightly more, the other. ■

In the simultaneous frame the invariance of the interval gives $-\Delta x'^2 = \Delta s^2$, so the two events are separated by

$$\Delta \sigma \equiv \sqrt{-\Delta s^2} = |\Delta x'|, \quad (2.3.30)$$

the **proper distance** — the spacelike counterpart of proper time, and the distance measured by the one observer for whom the two events are "at the same moment". Just as $\sqrt{\Delta s^2}$ is the smallest $c |\Delta t'|$ any observer can report for a timelike pair, $\sqrt{-\Delta s^2}$ is the smallest $|\Delta \mathbf{x}'|$ any observer can report for a spacelike pair, by the identical algebra.

Two events being reversible in order sounds alarming until you notice what it costs to arrange, and what it does not permit. Reversal requires $|\beta| > |\beta_0| = c |\Delta t| / |\Delta x|$, and that is possible *exactly* when the separation is spacelike. The condition for the reordering to be achievable and the condition for the events to be causally disconnected are the same condition. That is not luck. It is the theorem below.

4.5 · Why c is a causal speed limit and not merely light's speed

Assemble the two results. Time order is absolute for causally connectable pairs, and reversible for causally disconnected ones. Nothing so far forbids a faster-than-light influence; we simply have not allowed one. Allow one and watch.

Suppose some influence — a signal, a particle, a "quantum of whatever" — travels from event A to event B at speed $u > c$ in some frame. Then $|\Delta \mathbf{x}| = u |\Delta t| > c |\Delta t|$, so A and B are **spacelike separated**. By §4.4 there is an inertial frame in which B precedes A . But A was the cause and B the effect. So:

THE THEOREM

If any influence could travel faster than c , then some inertial observer would see the effect happen before the cause.

Not "would see it that way as an optical illusion" — would find, in coordinates constructed with their own synchronised clocks, that the effect is recorded at an earlier time than the cause. And by Postulate 1 that observer's description is as valid as anybody's.

An effect preceding its cause is uncomfortable but not yet a contradiction. The contradiction takes one more step, and the step is constructive: with two superluminal signals you can send a message into your own past.

The construction

Work with $c = 1$, in seconds and light-seconds, and take the idealised case of a signal that is instantaneous in the frame of whoever sends it. (The finite- u version is in the grind box; it needs $\beta > 2u/(u^2 + 1)$ and is otherwise identical.)

Step 1. Alice is at rest in frame S at $x = 0$. At $t = 0$ she sends a signal that is instantaneous in S — it travels along the line $t = 0$ — to Bob, who is at $x = L = 1$ light-second. The reception event is

$$P : (ct, x) = (0, 1). \quad (2.3.31)$$

Step 2. Bob is not at rest in S ; he moves at $\beta = 0.6$, so he is at rest in S' . In his coordinates, event P occurs at

$$\begin{aligned} ct'_P &= \gamma(ct_P - \beta x_P) = 1.25(0 - 0.6 \times 1) = -0.75, \\ x'_P &= \gamma(x_P - \beta ct_P) = 1.25(1 - 0) = 1.25. \end{aligned} \quad (2.3.32)$$

Step 3. On receiving the message, Bob immediately replies with a signal that is instantaneous in *his* frame — travelling along the line $t' = t'_P = -0.75$. Where does that line cross Alice's worldline $x = 0$? On $x = 0$ we have $ct' = \gamma ct$, so

$$\gamma ct = -0.75 \quad \implies \quad ct = -0.6. \quad (2.3.33)$$

Alice receives Bob's reply at $t = -0.6$ s: six tenths of a second *before* she sent the original message. In general the return time is $t = -\beta L/c$, so the effect is available at will and can be made as large as you like by increasing L .

Step 4, the contradiction. Alice now writes a program: *if a reply arrives before $t = 0$, do not send the message*. The premise of the loop is that the message was sent; the conclusion is that it was not. This is not a paradox of interpretation, a puzzle about free will, or a subtlety about what "cause" means. It is a straightforward logical contradiction produced by three ingredients: the existence of superluminal signalling, the relativity of simultaneity, and Postulate 1 (which grants Bob the same right as Alice to call his own frame's instantaneous signal instantaneous). Since the last two are established, the first must go.

WHAT c ACTUALLY IS

This is why the constant c is a **causal speed limit** and not merely "the speed of light". Nothing in the argument above mentioned light. The limit is a property of the causal structure of spacetime — of the light cone as the boundary between "can influence" and "cannot" — and light travels at c because light is massless, not the other way round.

The distinction is not pedantry, and it pays off twice later. If the photon turned out to have a tiny mass, light would travel slightly slower than c and not one line of this chapter would change; c would remain the limit and light would merely stop saturating it. And in general relativity, where there is no global speed to speak of, what survives is exactly this — the light cone at each point, the local causal structure, which Part III keeps while discarding almost everything else. That is why Chapter 3.1 can throw away flat spacetime and still have a theory.

▼ Grind box — the loop with a finite superluminal speed, and the exact threshold

The instantaneous case above is clean but looks like a limiting trick. It is not; any $u > c$ works, provided the relative velocity of the two parties is large enough. Here is the condition, derived.

Units $c = 1$. Alice at $x = 0$ sends at $t = 0$ a signal of speed $u > 1$ in S toward $+x$; Bob receives it at

$$P = (t_P, x_P) = (L/u, L).$$

Bob is at rest in S' with velocity β . Transform:

$$t'_P = \gamma L \left(\frac{1}{u} - \beta \right), \quad x'_P = \gamma L \left(1 - \frac{\beta}{u} \right).$$

Bob replies with a signal of speed u in *his own frame*, travelling in the $-x'$ direction: $x'(t') = x'_P - u(t' - t'_P)$. Alice's worldline $x = 0$ is, in S' , the line $x' = -\beta t'$. Setting the two equal and solving,

$$t'(u - \beta) = x'_P + u t'_P \quad \implies \quad t' = \frac{x'_P + u t'_P}{u - \beta}.$$

Substitute the two transformed coordinates and factor out γL :

$$x'_P + u t'_P = \gamma L \left[\left(1 - \frac{\beta}{u}\right) + (1 - \beta u) \right] = \gamma L \left[2 - \beta \left(u + \frac{1}{u}\right) \right].$$

Alice's own clock reads $t = t'/\gamma$ on her worldline, so the reply arrives at

$$t = \frac{L}{u - \beta} \left[2 - \beta \frac{u^2 + 1}{u} \right].$$

Since $u > \beta$, the sign is the sign of the bracket, and the reply beats the message ($t < 0$) exactly when

$$\boxed{\beta > \frac{2u}{u^2 + 1}.$$

Sanity checks. The threshold is less than 1 for every $u > 1$ — write it as $2u/(u^2 + 1) < 1 \Leftrightarrow (u - 1)^2 > 0$ ✓ — so an allowed β always exists. As $u \rightarrow \infty$ the threshold goes to 0: with truly instantaneous signals *any* relative motion suffices, matching the main text. As $u \rightarrow 1^+$ the threshold goes to 1: at exactly light speed you need an observer at exactly light speed, i.e. none, and causality is safe. The barrier is not a convention; it is precisely where the construction stops working.

Numbers. Take $u = 2c$, so the threshold is $2 \cdot 2/5 = 0.8$. With $L = 1$ light-second:

β	0.80	0.85	0.90	0.95
reply arrives at t (s)	0.000	−0.109	−0.227	−0.357

At threshold the reply arrives exactly as the message departs; beyond it, before.

One honest caveat. The argument shows that superluminal *signalling* — the transmission of information or influence — is inconsistent with relativity plus logic. It does not forbid superluminal coordinate speeds, which Chapter 2.2 §5 already showed are harmless (the spotlight, the scissors), and it does not by itself forbid a particle that always moves faster than c and cannot be used to signal. ▀ Whether such a particle is consistent quantum-mechanically is a separate question with a well-known answer — it is not — and the argument is Chapter 5.1's, not ours.

△ WHY THIS ISN'T OBVIOUS — ΔS^2 IS NOT A DISTANCE

Chapter 0.5 §1 laid down the axioms an inner product must satisfy, and one of them was *positive-definiteness*: $\langle v, v \rangle > 0$ for every $v \neq 0$. Everything downstream in that chapter leaned on it. The Cauchy–Schwarz inequality needs it. So does the triangle inequality, the definition of $\|v\| = \sqrt{\langle v, v \rangle}$ as a real number, the interpretation of $\langle u, v \rangle / \|u\| \|v\|$ as a cosine, and the whole apparatus of angles.

$\eta_{\mu\nu}$ **does not satisfy it, and the violation is deliberate.** The vector $(1, 1, 0, 0)$ is not zero, and $\eta_{\mu\nu} V^\mu V^\nu = 1 - 1 = 0$. So we have a nonzero vector of zero length, and vectors of negative length squared, and no way to define $\|V\|$ as a real number for all V . Calling η a "metric" is standard and is a slight abuse; the precise term is a *non-degenerate symmetric bilinear form of signature (1, 3)*, and the word non-degenerate marks the axiom we *did* keep.

What survives: bilinearity, symmetry, non-degeneracy (if $V \cdot W = 0$ for every W then $V = 0$), the invariance of $V \cdot W$ under the transformations, the notion of orthogonality $V \cdot W = 0$, and — crucially — the ability to classify vectors into three invariant types. That is enough to do geometry with, and §5 and §6 do.

What does not: positivity, the triangle inequality (§6 reverses it), Cauchy–Schwarz (§6 reverses that too, for timelike vectors), and the idea that "distance zero" means "same point". Vectors orthogonal to themselves exist and are exactly the null vectors — a sentence that would be simply false in Chapter 0.5 and is the defining feature of the light cone here.

So the honest summary is: we removed one axiom from the definition of an inner product, and in exchange the geometry acquired a causal structure. That trade is what the rest of physics is built on.

5 • Proper time is the length of a worldline

So far the interval has connected two events by a straight line. Now bend it.

A particle's history in spacetime is a curve, its **worldline**. Along it, take two neighbouring events separated by $dx^\mu = (c dt, dx, dy, dz)$. If the particle moves slower than light — which, by §4, is exactly the statement that its worldline is timelike at every point — then $ds^2 > 0$ and we may define

$$c^2 d\tau^2 \equiv ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2. \quad (2.3.34)$$

Divide the right-hand side by $c^2 dt^2$ and pull dt out:

$$d\tau = dt \sqrt{1 - \frac{1}{c^2} \left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 + \left(\frac{dz}{dt} \right)^2 \right]} = \sqrt{1 - \beta^2(t)} dt = \frac{dt}{\gamma(t)}, \quad (2.3.35)$$

where $\beta(t)$ is the particle's instantaneous speed in units of c — a function of t , because the particle may do whatever it likes. Integrate along the path:

$$\tau[\text{path}] = \int_{t_1}^{t_2} \sqrt{1 - \beta^2(t)} dt = \frac{1}{c} \int ds. \quad (2.3.36)$$

This quantity is the **proper time** along the worldline. Three things about it, in increasing order of importance.

It is invariant. Each ds is invariant, and a sum of invariants is invariant. Two observers computing (2.3.36) for the same path will use different $\beta(t)$, different t_1, t_2 , and different intermediate values, and will get the same number.

It is a functional of the path, not a function of the endpoints. This is the language of Chapter 1.2: τ takes an entire curve as input and returns one number, in precisely the way that the arc-length functional $\mathcal{A}[y] = \int \sqrt{1 + y'^2} dx$ of 1.2 §1 does. Compare them directly —

$$\underbrace{\mathcal{A}[y] = \int \sqrt{dx^2 + dy^2}}_{\text{Euclidean length}}, \quad \underbrace{c \tau[\text{path}] = \int \sqrt{c^2 dt^2 - dx^2}}_{\text{Minkowski length}} \quad (2.3.37)$$

— and there is the sign again, in the only place it could possibly be. **Proper time is arc length.** It is the length of the worldline, measured with the geometry that spacetime actually has. Everything §6 does follows from taking that sentence at face value.

It is what a clock reads. This is the part that keeps proper time from being an abstraction. Take the interval between two neighbouring events on a clock's own worldline. In the clock's instantaneous rest frame it is not moving, so $dx' = dy' = dz' = 0$ and $ds^2 = c^2 dt'^2$ — the interval is just the time elapsed on that clock. Since ds^2 is invariant, the same is true computed in any frame. So τ is the accumulated reading of a clock carried along the path, and (2.3.36) says a moving clock accumulates less of it than the coordinate time, by exactly γ at each instant — which is Chapter 2.2's time dilation, now derived as a statement about the length of a curve rather than about clocks being defective.

▼ Grind box — the assumption hidden in "a clock measures proper time"

The argument above computed $d\tau$ in the clock's *instantaneous* rest frame, and then added the pieces up. There is an assumption buried in "instantaneous", and it deserves naming because it is a physical input rather than a mathematical one.

The clock hypothesis. *An ideal clock's rate depends only on its instantaneous velocity, not on its acceleration.* Equivalently: an accelerating clock reads the same as a momentarily co-moving inertial clock, at every moment.

This is not a theorem of special relativity. It is an extra assumption, and it is easy to see that it could fail: a pendulum clock in an accelerating rocket manifestly does not obey it, and neither does any device whose mechanism is disturbed by being shaken. What the hypothesis really asserts is that *some* physical processes are ideal in this sense, so that "proper time" is realisable and not merely definable.

■ Experimentally it holds to an astonishing degree. Muons circulating in a storage ring at $\gamma \approx 29$ experience a proper acceleration of order $10^{18}g$, and their decay rate is slowed by exactly the factor γ predicted from their speed alone, with no acceleration-dependent correction at the 10^{-3} level. Atomic clocks flown on aircraft and the clocks aboard GPS satellites agree with (2.3.36) integrated along their actual trajectories. We quote these; deriving them would require the internal dynamics of the clock.

Why it matters twice. First, without it, §6's twin calculation would be incomplete — you could always object that the turnaround does something unmodelled to the traveller's clock. Second, in general relativity the clock hypothesis is promoted to a postulate: proper time along a worldline is *defined* to be what a clock carried along it reads, and that definition is the bridge between the geometry and any measurement at all. Chapter 3.3 leans on it in its first paragraph.

6 · The twin paradox is a theorem about triangles

Now the payoff. In the plane, the straight line is the shortest path between two points, and everyone has known that since childhood. In spacetime the corresponding theorem is true with the inequality turned around, and the reversal is the minus sign, doing its most consequential work.

6.1 · The theorem

STRAIGHT IS LONGEST

Among all timelike worldlines connecting two fixed timelike-separated events, the **straight one has the greatest proper time**. Every other path takes strictly less, and the more it deviates, the less it takes.

Proof. Let the two events be E_1 and E_2 , timelike separated. By §4.3 and the invariance of the interval there is a frame in which they occur at the same place — take $\beta_{\text{rest}} = \Delta x / (c\Delta t)$, which satisfies $|\beta_{\text{rest}}| < 1$ precisely because the separation is timelike, exactly mirroring the construction in §4.4. Work in that frame. There, $\Delta \mathbf{x} = 0$ and the coordinate time between the events is $T = \Delta t$, which by (2.3.27) is $\sqrt{\Delta s^2}/c$.

The straight worldline joining them is the one at rest in this frame. Along it $\beta \equiv 0$, so (2.3.36) gives $\tau_{\text{straight}} = T$.

Now take *any* timelike path between the same two events. It has the same endpoints, hence runs over the same coordinate-time interval, and

$$\tau[\text{path}] = \int_0^T \sqrt{1 - \beta^2(t)} \, dt \leq \int_0^T 1 \, dt = T = \tau_{\text{straight}}, \quad (2.3.38)$$

because $\sqrt{1 - \beta^2} \leq 1$ for every β , with equality only at $\beta = 0$. So every path loses, and it loses strictly unless it is at rest for the entire journey — that is, unless it is the straight one. ■

That is the whole proof. It is four lines because we chose the right frame first, and we were allowed to choose it because the answer is frame-independent. The integrand $\sqrt{1 - \beta^2}$ is a penalty for moving, charged continuously, and the straight path is simply the one that never pays it.

6.2 · The same result from Chapter 1.2's machinery

The direct argument is airtight but it uses a special frame. The variational argument does not, and it is the one that survives into Part III, so it is worth doing even though we already have the answer.

Extremise the functional (2.3.36). In one space dimension, with $\dot{x} = dx/dt$,

$$\tau[x] = \int_{t_1}^{t_2} F(\dot{x}) dt, \quad F(\dot{x}) = \sqrt{1 - \frac{\dot{x}^2}{c^2}}. \quad (2.3.39)$$

This is a Chapter 1.2 problem of exactly the type solved in its worked example 2. The Euler–Lagrange equation is

$$\frac{d}{dt} \left(\frac{\partial F}{\partial \dot{x}} \right) - \frac{\partial F}{\partial x} = 0, \quad (2.3.40)$$

and since F contains no x at all — spacetime is homogeneous; there is no preferred place — the second term vanishes and the first bracket is a constant of the motion:

$$\frac{\partial F}{\partial \dot{x}} = \frac{-\dot{x}/c^2}{\sqrt{1 - \dot{x}^2/c^2}} = \text{const} \equiv -\frac{K}{c^2}. \quad (2.3.41)$$

Solve for $\dot{x}$ exactly as 1.2 did for the plane geodesic: square, clear the denominator, and collect,

$$\begin{aligned} \dot{x}^2 = K^2 \left(1 - \frac{\dot{x}^2}{c^2} \right) &\implies \dot{x}^2 \left(1 + \frac{K^2}{c^2} \right) = K^2 \\ &\implies \dot{x} = \frac{K}{\sqrt{1 + K^2/c^2}} = \text{const}. \end{aligned} \quad (2.3.42)$$

Constant velocity: $x = vt + x_0$, a **straight worldline**. We assumed nothing about the answer, only stationarity. And note the free bonus in the last expression — whatever real constant K you pick, the resulting $|\dot{x}|$ is strictly less than c . The extremal paths are automatically timelike; the variational principle cannot produce a superluminal answer even if you ask it to.

The variational calculation identifies the straight line as *stationary*. Which kind of stationary point it is comes from §6.1: a maximum. Chapter 1.2 §5.1 made the same distinction for the action and warned that stationary need not mean minimal; here is the cleanest possible example of the warning being necessary.

WHERE THIS GOES, AND IT IS THE MOST IMPORTANT FORWARD POINTER IN PART II

Look at what the calculation actually needed. It needed a functional of the form $\int ds$ with $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$, and the Euler–Lagrange machinery of Chapter 1.2. It did not need flatness, it did not need η to be constant, and it did not need Lorentz transformations.

Chapter 3.3 will keep this variational principle *word for word* and make exactly one change: replace the constant array $\eta_{\mu\nu}$ by a position-dependent one, $g_{\mu\nu}(x)$ — ten functions of where you are. The same functional, the same Euler–Lagrange equation, the same request for a stationary path. The resulting equation is called the **geodesic equation**, and its solutions are called *free fall*.

That is the whole of general relativity's kinematics, and this section is where it starts to be earned. Chapter 1.2 already showed you the pattern once: the same $\int ds$ gave a straight line in the plane and a great circle on a sphere, and the only thing that changed was ds . Gravity is that substitution performed on spacetime. A planet orbits the Sun for the same reason a great-circle route from London to Tokyo goes over the Arctic — it is going as straight as the geometry allows.

6.3 · The reverse triangle inequality

The theorem of §6.1 has a clean algebraic form worth stating on its own, because it is the exact mirror image of the most familiar inequality in geometry.

Call a four-vector V^μ *timelike* if $V \cdot V > 0$, and *future-pointing* if $V^0 > 0$; for such vectors write $|V| \equiv \sqrt{V \cdot V}$, which is a real number.

$$\text{For future-pointing timelike } U, V : \quad |U + V| \geq |U| + |V|, \quad (2.3.43)$$

with equality only when U and V are parallel. Compare the Euclidean statement $\|u + v\| \leq \|u\| + \|v\|$: the inequality has flipped. In the plane, going via a third point makes the journey longer; in spacetime it makes the elapsed time shorter, and the two-leg twin path is precisely $U + V$ with U the outbound leg and V the inbound one. Problem 1 asks you to prove (2.3.43), and the proof turns on a reversed Cauchy–Schwarz inequality $U \cdot V \geq |U| |V|$ — also flipped, and flipped for the same reason.

6.4 · Closing the loop with Chapter 2.2

Chapter 2.2 §7.1 worked the twin paradox in detail: traveller goes 4 light-years at $\beta = 0.8$, turns, returns; Earth ages 10 years, the traveller ages 6. The resolution offered there was that the traveller changes inertial frames and the stay-at-home does not, followed by a careful accounting of the 6.4 years that the traveller's simultaneity slice sweeps across Earth's worldline during the turnaround. All correct, and all now unnecessary.

Here is the same result in one sentence. **The travelling twin ages less because a bent timelike path is shorter, and shorter means less proper time.**

$$\tau_{\text{straight}} = \sqrt{10^2 - 0^2} = 10 \text{ yr}, \quad \tau_{\text{bent}} = 2\sqrt{5^2 - 4^2} = 2 \times 3 = 6 \text{ yr}. \quad (2.3.44)$$

Two paths, same endpoints, different lengths. Nobody is scandalised that a detour adds mileage; the only unfamiliar thing is that in this geometry a detour *subtracts* time. Notice what the geometric statement does not require: it never mentions acceleration, never uses a non-inertial frame, and never needs the traveller's point of view at all. Acceleration enters only as the answer to a different question — *why is one path bent?* — and a kink in a worldline is detectable from the inside, by an accelerometer, which is why there was never a symmetry between the twins to appeal to.

The often-heard slogan "the twin paradox is really about acceleration" is therefore half wrong. Acceleration is what makes a path bent; bentness is what costs proper time; and the amount of time lost is not a function of the acceleration but of the whole shape of the path. Worked example 2 makes that concrete by comparing three paths between the same two events, one of them smoothly accelerated throughout with no kink anywhere.

▼ Grind box — the second variation, and why the extremum is a maximum rather than a minimum

Section 6.1 proved the maximum by a global argument. Here is the local version, which is what Chapter 1.2 §5.1's second-variation machinery would say, and it exposes the sign in a different place.

Work in the frame where the endpoints coincide spatially, and perturb the straight path $x(t) = 0$ by a small $\varepsilon \xi(t)$ with $\xi(t_1) = \xi(t_2) = 0$. Then $\beta = \varepsilon \dot{\xi}/c$ and

$$\tau[\varepsilon] = \int \sqrt{1 - \frac{\varepsilon^2 \dot{\xi}^2}{c^2}} dt = \int \left[1 - \frac{\varepsilon^2 \dot{\xi}^2}{2c^2} - \frac{\varepsilon^4 \dot{\xi}^4}{8c^4} - \dots \right] dt,$$

by the binomial series of Chapter 0.3. Read off the two variations:

$$\delta\tau = 0 \quad (\text{no term linear in } \varepsilon), \quad \delta^2\tau = -\frac{\varepsilon^2}{c^2} \int \dot{\xi}^2 dt < 0.$$

The first variation vanishes for *every* perturbation, confirming stationarity without solving anything. The second variation is **negative definite** — strictly negative unless $\dot{\xi} \equiv 0$, i.e. unless the perturbation is nothing at all. Hence a maximum, and a genuine one, not a saddle.

Where the sign came from. Compare the Euclidean calculation for a straight line in the plane, perturbed the same way: $\int \sqrt{1 + \varepsilon^2 \xi'^2} dx = \int [1 + \frac{1}{2}\varepsilon^2 \xi'^2 + \dots]$, giving $\delta^2\mathcal{A} = +\varepsilon^2 \int \xi'^2 > 0$: a minimum. The two calculations are identical except for the sign inside the square root, which is the sign in the metric, which flips the sign of the second variation, which flips minimum into maximum. One symbol, all the way through.

And a warning that will matter in Part III. This is a *local* statement. It says the straight path beats all nearby paths. In flat spacetime the global statement is also true — that is §6.1 — but in curved spacetime it can fail: there can be several geodesics between the same two events, and only one of them maximises. That is the conjugate-point phenomenon of Chapter 1.2 §5.1, and in Chapter 3.7 the several geodesics become the several images of a gravitationally lensed quasar.

7 · Four-velocity, briefly

One construction before we stop, because it is short, it is forced, and Chapter 2.5 is built entirely on it.

Ordinary velocity is $d\mathbf{x}/dt$, and Chapter 2.2's Problem 1 exposed its defect: the numerator is a piece of a four-vector but the denominator is a coordinate, so the quotient transforms in a mess. Both the top and the bottom change when you change frames, and they do not change compatibly. The repair is to divide by something invariant, and §5 has just supplied the only natural candidate. Define the **four-velocity**

$$u^\mu \equiv \frac{dx^\mu}{d\tau}. \quad (2.3.45)$$

Since dx^μ transforms as a four-vector and $d\tau$ is a number every observer agrees on, u^μ is a four-vector — the numerator picks up a factor of Λ and the denominator picks up nothing. That is the entire reason for the definition.

Its components follow from $d\tau = dt/\gamma$:

$$u^\mu = \frac{dt}{d\tau} \frac{dx^\mu}{dt} = \gamma (c, \mathbf{v}) = (\gamma c, \gamma \mathbf{v}). \quad (2.3.46)$$

And now the property that makes it the right object. Compute its invariant square:

$$u \cdot u = \eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = \frac{\eta_{\mu\nu} dx^\mu dx^\nu}{d\tau^2} = \frac{ds^2}{d\tau^2} = c^2. \quad (2.3.47)$$

Identically — for every particle, at every moment, in every frame, whatever it is doing. Check it against the components: $\gamma^2 c^2 - \gamma^2 v^2 = \gamma^2 c^2 (1 - \beta^2) = c^2 \checkmark$.

AN OBJECT OF FIXED LENGTH

The four-velocity is a four-vector of **constant Minkowski length** c . It can point in any timelike future direction, and that is all the freedom it has — it lives on the invariant hyperbola of §3, which in four dimensions is a three-dimensional hyperboloid, and boosting a particle slides its four-velocity along that surface exactly as the figure in §3.4 slides an event along a hyperbola.

This is why u^μ and not $\mathbf{v}$ is the right generalisation of velocity: the constraint $u \cdot u = c^2$ is frame-independent, whereas "speed less than c " is a statement about components. Chapter 2.5 multiplies by the rest mass to get $p^\mu = m u^\mu$, whose invariant square is then $p \cdot p = m^2 c^2$ with no work at all, and that single equation contains $E = \gamma m c^2$, $E^2 = p^2 c^2 + m^2 c^4$, and the massless case $E = pc$ as its three readings.

△ THE SLOGAN, AND WHAT IS ACTUALLY TRUE

You will hear it said that *everything moves through spacetime at speed c* , and that time dilation is what happens when motion through space is "bought" out of motion through time. As a mnemonic it is not bad. As a derivation it is not one, and it is worth separating the two.

True: $u \cdot u = c^2$ for every particle, by (2.3.47). That is a real theorem and it is what the slogan is gesturing at.

Careful: the components of u^μ are $(\gamma c, \gamma \mathbf{v})$, and these do *not* trade off the way a budget does. Writing the invariant out,

$$(u^0)^2 = c^2 + |\mathbf{u}|^2,$$

so as spatial motion grows, the "motion through time" $u^0 = \gamma c$ **grows too** — it does not shrink. There is no fixed pot of c being divided; the two pieces add with a minus sign, which is precisely why the total can stay at c while both pieces run off to infinity. A picture built on "you only have so much speed to share out" will give you the wrong sign on the first question that tests it.

The honest version of the slogan: a particle's four-velocity is a unit timelike vector (scaled by c), and its direction — not its magnitude — encodes the velocity. Time dilation is the statement that a unit vector tilted away from your time axis has a smaller projection onto it, and the projection is $d\tau/dt = 1/\gamma$. That is a statement about a fixed-length vector being tilted, which is exactly §3's picture, and it is derivable. The budget metaphor is not.

8 · Worked examples

WORKED EXAMPLE 1 — A FULL CAUSAL-STRUCTURE COMPUTATION

Four events are given in a frame S , in units where $c = 1$: time in years, distance in light-years, so coordinates are written (ct, x) with $y = z = 0$ throughout.

EVENT	P_1	P_2	P_3	P_4
(ct, x)	$(0, 0)$	$(3, 1)$	$(1, 4)$	$(5, 5)$

(a) Classify all six pairs. (b) For the spacelike pair P_2P_3 , find the frame in which they are simultaneous and a frame in which their order is reversed, and verify the interval is unchanged. (c) Show that no boost reverses the timelike pair P_1P_2 , and find the frame that comes closest.

(a) The classification. Compute $\Delta s^2 = c^2\Delta t^2 - \Delta x^2$ for each pair. Six pairs, six subtractions:

PAIR	$c\Delta t$	Δx	Δs^2	TYPE	INVARIANT MEANING
$P_1 \rightarrow P_2$	3	1	$9 - 1 = +8$	timelike	$\tau = \sqrt{8} = 2.828 \text{ yr}$
$P_1 \rightarrow P_3$	1	4	$1 - 16 = -15$	spacelike	$\sigma = \sqrt{15} = 3.873 \text{ ly}$
$P_1 \rightarrow P_4$	5	5	$25 - 25 = 0$	null	on the light cone
$P_2 \rightarrow P_3$	-2	3	$4 - 9 = -5$	spacelike	$\sigma = \sqrt{5} = 2.236 \text{ ly}$
$P_2 \rightarrow P_4$	2	4	$4 - 16 = -12$	spacelike	$\sigma = \sqrt{12} = 3.464 \text{ ly}$
$P_3 \rightarrow P_4$	4	1	$16 - 1 = +15$	timelike	$\tau = \sqrt{15} = 3.873 \text{ yr}$

Read the structure off. P_2 and P_4 both lie in the absolute future of P_1 's light cone or on it — P_2 strictly inside, P_4 exactly on it, meaning a light signal emitted at P_1 arrives precisely at P_4 . P_3 is in P_1 's *elsewhere*: nothing that happens at P_1 can affect P_3 , and vice versa. Yet P_3 can affect P_4 ($\Delta s^2 = +15$), and so can P_1 . Causal influence is not a transitive-and-total ordering; it is a partial order, and the light cones are its structure.

Note also that the classification survives translation: only differences entered, so moving the origin changes nothing. That is Chapter 2.2's homogeneity, still holding.

(b) The spacelike pair P_2P_3 . In S , P_3 happens 2 years *before* P_2 ($c\Delta t = -2$ going from P_2 to P_3) and 3 light-years to the right.

Simultaneous frame. By (2.3.29),

$$\beta_0 = \frac{c \Delta t}{\Delta x} = \frac{-2}{3} = -0.6667,$$

which has modulus less than 1 ✓, as guaranteed for a spacelike pair. That frame moves in the $-x$ direction at two thirds of light speed. In it, $\gamma_0 = (1 - 4/9)^{-1/2} = 1.3416$ and

$$\begin{aligned}\Delta x' &= \gamma_0(\Delta x - \beta_0 c \Delta t) \\ &= 1.3416(3 - (-0.6667)(-2)) = 1.3416 \times 1.6667 = 2.2361,\end{aligned}$$

and $2.2361 = \sqrt{5}$ ✓ — the proper distance, exactly as (2.3.30) promised. The two events are $\sqrt{5}$ light-years apart and simultaneous, for that observer.

Reversed frame. Any β beyond β_0 in the same direction does it; take $\beta = -0.8, \gamma = 5/3$:

$$\begin{aligned}c \Delta t' &= \frac{5}{3}(-2 - (-0.8)(3)) = \frac{5}{3}(0.4) = +0.6667, \\ \Delta x' &= \frac{5}{3}(3 - (-0.8)(-2)) = \frac{5}{3}(1.4) = 2.3333.\end{aligned}$$

The sign of $\Delta t'$ has flipped: in S , P_3 came first; in this frame P_2 does. And the interval is untouched:

$$\Delta s'^2 = (0.6667)^2 - (2.3333)^2 = 0.4444 - 5.4444 = -5.0000 \quad \checkmark$$

Both frames are correct. Neither event caused the other — they cannot, being spacelike separated — so nothing whatever is at stake in the disagreement.

(c) The timelike pair P_1P_2 , which no boost can reorder. Here $c \Delta t = 3, \Delta x = 1$. From (2.3.25),

$$c \Delta t' = \gamma(3 - \beta),$$

and since $|\beta| < 1 < 3$, the bracket is positive for every admissible β : at worst $\beta \rightarrow 1^-$ gives $3 - \beta \rightarrow 2$, still positive, while $\gamma \rightarrow \infty$. So $\Delta t' > 0$ always, and in fact $\Delta t'$ can be made arbitrarily *large* but never small. Tabulate:

β	-0.9	0	+1/3	+0.8	+0.99
$c \Delta t'$ (yr)	8.947	3.000	2.828	3.667	14.25
$\Delta x'$ (ly)	8.488	1.000	0	-2.333	-13.96
$\Delta s'^2$	8.000	8.000	8.000	8.000	8.000

The last row is the point of the whole exercise: two coordinates swinging over an order of magnitude and changing sign, and one number sitting perfectly still.

The closest approach. The minimum of $c \Delta t'$ is at $\beta = \Delta x / (c \Delta t) = 1/3$, the frame in which the two events happen at the same place, and there

$$c \Delta t'_{\min} = \sqrt{\Delta s^2} = \sqrt{8} = 2.8284 \text{ yr}, \quad \Delta x' = 0,$$

confirming (2.3.27). A clock that is present at both events — it can be, since they are timelike separated; it needs to travel 1 light-year in 3 years, i.e. at $\beta = 1/3$ — reads 2.828 years between them, and that is the proper time of the pair. No observer can shrink the gap below it, and none can make the two events simultaneous, let alone reversed.

The moral. Time order is not "relative". It is *absolute exactly where it could matter and relative exactly where it could not*, and the boundary between the two regimes is the light cone. That is a far more disciplined statement than "everything is relative", and it is the one the theory actually makes.

WORKED EXAMPLE 2 — THREE WORLDLINES BETWEEN THE SAME TWO EVENTS

Two events: departure $E_1 = (ct, x) = (0, 0)$ and reunion $E_2 = (10, 0)$, again in years and light-years with $c = 1$. Compute the proper time along three timelike paths joining them: (a) the straight one; (b) Chapter 2.2's two-leg trip out to $x = 4$ at $\beta = 0.8$ and back; (c) a smooth worldline of constant proper acceleration with the same turning point. Verify that the straight one wins.

(a) Straight. The path $x(t) = 0$ has $\beta \equiv 0$, so (2.3.36) gives

$$\tau_a = \int_0^{10} \sqrt{1 - 0} \, dt = 10.000 \text{ yr.}$$

Equivalently, straight from the interval: $\tau = \sqrt{\Delta s^2}/c = \sqrt{10^2 - 0^2} = 10$.

(b) Two straight legs. Outbound from $(0, 0)$ to $(5, 4)$, inbound from $(5, 4)$ to $(10, 0)$. Each leg is straight, so its proper time is the interval along it:

$$\tau_b = \sqrt{5^2 - 4^2} + \sqrt{5^2 - 4^2} = 3 + 3 = 6.000 \text{ yr.}$$

Speed on each leg: $\beta = 4/5 = 0.8$, so $\sqrt{1 - \beta^2} = 0.6$ and $5 \times 0.6 = 3 \checkmark$ by the other route. This is Chapter 2.2 §7.1's traveller, and the deficit is $10 - 6 = 4$ years.

(c) Constant proper acceleration. Chapter 2.2's Problem 4 showed that a worldline of constant proper acceleration is a hyperbola $(x - x_c)^2 - c^2(t - t_c)^2 = k^2$ — which §3 now identifies as an *invariant hyperbola*, translated. Take the branch that leaves $x = 0$, turns around, and comes back, symmetric about $ct = 5$:

$$x(t) = x_c - \sqrt{k^2 + c^2(t - 5)^2}, \quad x_c = \sqrt{k^2 + 25},$$

where x_c was fixed by demanding $x(0) = 0$, and $x(10) = 0$ then follows by symmetry. Choose k so that the turning point matches path (b)'s, namely $x(5) = x_c - k = 4$:

$$\sqrt{k^2 + 25} - k = 4 \implies k^2 + 25 = k^2 + 8k + 16 \implies k = \frac{9}{8} = 1.125 \text{ ly.}$$

Is it timelike everywhere? Write $u = c(t - 5)$. Then $dx/du = -u/\sqrt{k^2 + u^2}$, whose modulus is less than 1 for every $u \checkmark$. The fastest it ever goes is at the endpoints, $|\beta|_{\max} = 5/\sqrt{1.125^2 + 25} = 0.9756$.

Proper time. Along the path,

$$c^2 d\tau^2 = du^2 - dx^2 = du^2 \left[1 - \frac{u^2}{k^2 + u^2} \right] = \frac{k^2 du^2}{k^2 + u^2},$$

so $c \, d\tau = k \, du / \sqrt{k^2 + u^2}$, which integrates to an inverse hyperbolic sine:

$$c \, \tau_c = k \left[\operatorname{arsinh} \frac{u}{k} \right]_{-5}^{+5} = 2k \operatorname{arsinh} \frac{5}{k} = 2(1.125) \operatorname{arsinh}(4.4444).$$

Since $\operatorname{arsinh} z = \ln(z + \sqrt{1 + z^2})$, we get $\operatorname{arsinh}(4.4444) = \ln 9 = 2.19722$ (the square root comes out to exactly 4.5556, so the argument is exactly 9), hence

$$\tau_c = 2.25 \times 2.19722 = 4.9438 \text{ yr.}$$

(Checked by direct numerical summation of $\sqrt{dt^2 - dx^2}$ over two million steps along the path: 4.94375530, against the closed form 4.94375530.)

The comparison.

PATH	PROPER TIME	DEFICIT	LENGTH ON THE PAGE
(a) straight	10.0000 yr	—	10.00
(b) two legs, $\beta = 0.8$	6.0000 yr	4.00 yr	$2\sqrt{41} = 12.81$
(c) constant proper acceleration	4.9438 yr	5.06 yr	12.97

The straight path wins, as §6.1 guarantees it must. And the ordering of the last column is exactly the reverse of the second — more page-length, less proper time, monotonically — which is the $\triangle$ callout of §3.3 in numerical form.

Two things worth extracting.

Acceleration is not the mechanism. Path (c) has no kink anywhere; it is smooth, its proper acceleration is constant, and it never changes inertial frame abruptly. It nonetheless loses more proper time than path (b), which does have a kink. So "the twin who accelerates ages less" is not a law — the amount lost depends on the whole shape of the path, not on where or how hard it accelerated. The correct statement remains the geometric one: bent is shorter, and (c) is bent more.

The numbers are not exotic. The proper acceleration of path (c) is $a = c^2/k$, and Chapter 2.2's Problem 4 gives $c^2/g = 0.9687$ light-years, so

$$\frac{a}{g} = \frac{c^2/g}{k} = \frac{0.9687}{1.125} = 0.861.$$

A crew accelerating at $0.86g$ — slightly less than standing on Earth — makes a round trip that takes ten years by Earth's clocks and four years eleven months by theirs, reaching 4 light-years out and 97.6% of light speed at the extremes. This is not a thought experiment about impossible machines. It is a comfortable ride, and the only obstacle is fuel, which is Chapter 2.5's problem.

9 • Your turn

PROBLEM 1 — THE REVERSE TRIANGLE INEQUALITY

Let U^μ and V^μ be future-pointing timelike four-vectors: $U \cdot U > 0$, $V \cdot V > 0$, $U^0 > 0$, $V^0 > 0$. Write $|U| = \sqrt{U \cdot U}$. (a) Prove the *reversed* Cauchy–Schwarz inequality $U \cdot V \geq |U| |V|$, with equality only when U and V are parallel. (b) Deduce $|U + V| \geq |U| + |V|$. (c) Explain in one sentence which Euclidean fact each of these is the mirror image of, and where the sign flip entered.

– Solution

(a) Reversed Cauchy–Schwarz. Both sides of the claimed inequality are invariant, so we may evaluate them in any convenient frame — this is the standard and enormously useful move, and it is licensed by (2.3.4). Choose the rest frame of U : since U is timelike and future-pointing, §6.1's construction supplies a boost making its spatial part vanish, so

$$U^\mu = (|U|, \mathbf{0}),$$

because then $U \cdot U = (U^0)^2 = |U|^2$ and $U^0 > 0$. In that frame write $V^\mu = (V^0, \mathbf{V})$. The dot product is easy:

$$U \cdot V = \eta_{\mu\nu} U^\mu V^\nu = |U| V^0.$$

Now use V 's own invariant, $V \cdot V = (V^0)^2 - |\mathbf{V}|^2 = |V|^2$, to write $V^0 = \sqrt{|\mathbf{V}|^2 + |V|^2}$ — positive root, since V is future-pointing and, by the causality theorem of §4.3, that is frame-independent. Hence

$$U \cdot V = |U| \sqrt{|\mathbf{V}|^2 + |V|^2} \geq |U| |V|,$$

with equality if and only if $\mathbf{V} = \mathbf{0}$, i.e. V is at rest in U 's frame, i.e. $V \parallel U$. ■

The whole reversal happened in one place: $|\mathbf{V}|^2$ enters V^0 with a *plus* sign after the minus in the metric has been moved to the other side. In Euclidean space the corresponding step subtracts and you get $\leq$.

(b) The triangle inequality. $U + V$ is future-pointing (its time component is a sum of positives) and timelike (by (a): $(U + V) \cdot (U + V) = |U|^2 + 2U \cdot V + |V|^2 > 0$), so $|U + V|$ is defined. Expand and apply (a):

$$|U + V|^2 = |U|^2 + 2U \cdot V + |V|^2 \geq |U|^2 + 2|U||V| + |V|^2 = (|U| + |V|)^2.$$

Take square roots — legitimate, both sides positive — to get $|U + V| \geq |U| + |V|$, with equality only for parallel vectors. ■

(c) The mirrors. (a) mirrors $|\langle u, v \rangle| \leq \|u\| \|v\|$ of Chapter 0.5 §1, and (b) mirrors $\|u + v\| \leq \|u\| + \|v\|$. Both flip because the metric is not positive-definite: in Chapter 0.5 the proof of Cauchy–Schwarz went by demanding $\langle u - \lambda v, u - \lambda v \rangle \geq 0$ for all λ , and that step is exactly the axiom §4's $\triangle$ callout removed.

What it says physically. Take U and V to be the two legs of the twin's journey, as four-vectors from departure to turnaround and turnaround to reunion. Then $U + V$ is the straight path, $|U + V| = c \times$ (stay-at-home's proper time) and $|U| + |V| = c \times$ (traveller's). The inequality is the twin paradox, and it is a two-line theorem about vectors. Numerically, with $U = (5, 4)$ and $V = (5, -4)$ in the units of §6.4: $|U| = |V| = 3$, $U + V = (10, 0)$, $|U + V| = 10 \geq 6 \checkmark$, and $U \cdot V = 25 + 16 = 41 \geq 3 \times 3 = 9 \checkmark$ — with plenty to spare, because the two legs are far from parallel.

PROBLEM 2 — SIMULTANEITY AND REVERSAL FOR A SPACELIKE PAIR

Two events in a frame S , in years and light-years with $c = 1$: $E_1 = (ct, x) = (0, 0)$ and $E_2 = (2, 5)$. (a) Classify the pair. (b) Find the frame in which they are simultaneous, and the distance between them in that frame. (c) Find a frame in which E_2 precedes E_1 , and verify the interval. (d) Show that no frame can make them occur at the same place, and say why that is the mirror image of the timelike case.

– Solution

(a) $\Delta s^2 = 2^2 - 5^2 = 4 - 25 = -21 < 0$: **spacelike**. Nothing at E_1 can influence E_2 ; a signal would need speed $5/2 = 2.5c$.

(b) **Simultaneous frame**. Set $\Delta t' = 0$ in $c\Delta t' = \gamma(c\Delta t - \beta\Delta x)$:

$$\beta_0 = \frac{c\Delta t}{\Delta x} = \frac{2}{5} = 0.4, \quad \gamma_0 = \frac{1}{\sqrt{1 - 0.16}} = 1.09109.$$

Then

$$\Delta x' = \gamma_0(\Delta x - \beta_0 c\Delta t) = 1.09109(5 - 0.8) = 1.09109 \times 4.2 = 4.58258,$$

and $4.58258 = \sqrt{21} \checkmark$, the proper distance (2.3.30). Check the interval: $0^2 - (\sqrt{21})^2 = -21 \checkmark$. An observer moving at $0.4c$ says these two events happen at the same moment, 4.583 light-years apart.

(c) **Reversed frame**. Any $\beta > 0.4$ works. Take $\beta = 0.8$, $\gamma = 5/3$:

$$c\Delta t' = \frac{5}{3}(2 - 0.8 \times 5) = \frac{5}{3}(-2) = -3.3333, \quad \Delta x' = \frac{5}{3}(5 - 0.8 \times 2) = \frac{5}{3}(3.4) = 5.6667.$$

$$\Delta s'^2 = (-3.3333)^2 - (5.6667)^2 = 11.1111 - 32.1111 = -21.0000 \quad \checkmark$$

In S , E_1 came first by 2 years; in this frame E_2 comes first by 3.33 years. Both correct, nothing at stake.

(d) **No common-place frame**. Setting $\Delta x' = 0$ would need $\beta = \Delta x/(c\Delta t) = 5/2 = 2.5$, which exceeds 1 and is not a velocity. More robustly, from invariance $|\Delta \mathbf{x}'|^2 = c^2 \Delta t'^2 + 21 \geq 21 > 0$ in every frame, so $\Delta \mathbf{x}'$ can never vanish and in fact can never fall below $\sqrt{21}$.

That is the exact mirror of the timelike result (2.3.27), where $c|\Delta t'| \geq \sqrt{\Delta s^2}$ and no frame can make the events simultaneous. Swap the roles of space and time and the two statements map onto each other, which is what you should expect of a geometry whose only asymmetry between them is a sign. In slogan form: **a timelike pair has an invariant time and a negotiable separation; a spacelike pair has an invariant separation and a negotiable time.**

PROBLEM 3 — THE ACCELERATED WORLDLINE IS A HYPERBOLA, AND IT HAS A HORIZON

A rocket has constant proper acceleration a . Chapter 2.2's Problem 4 gave its rapidity as $\phi = a\tau/c$ and its worldline, in the frame where it is momentarily at rest at $\tau = 0$, as

$$ct(\tau) = \frac{c^2}{a} \sinh \frac{a\tau}{c}, \quad x(\tau) = \frac{c^2}{a} \cosh \frac{a\tau}{c}.$$

(a) Show that this is an invariant hyperbola of §3 and identify which one. (b) Find its asymptotes. (c) A beacon sits at the spatial origin $x = 0$ and flashes at times t_0 . Show that flashes with $t_0 \geq 0$ never reach the rocket, and that flashes with $t_0 < 0$ always do. (d) Identify the resulting **Rindler horizon** and say precisely what it is a horizon for.

– Solution

(a) It is an invariant hyperbola. Use $\cosh^2 - \sinh^2 = 1$:

$$x^2 - c^2 t^2 = \frac{c^4}{a^2} \left[\cosh^2 \frac{a\tau}{c} - \sinh^2 \frac{a\tau}{c} \right] = \left(\frac{c^2}{a} \right)^2,$$

a constant. So the worldline lies on $c^2 t^2 - x^2 = -(c^2/a)^2$ — one of §3's *spacelike* hyperbolae, the right-hand branch, with $k = c^2/a$. Write $b \equiv c^2/a$ for brevity; it is a length, and it is the rocket's distance from the origin at $t = 0$.

This is a striking fact worth pausing on. A boost slides points along these curves ((2.3.19)), so **the uniformly accelerated worldline is an orbit of the boost group**. Boosting the rocket by ϕ gives you the same worldline, reparametrised. That is the exact analogue of "a circle is carried to itself by rotation", and it is why this trajectory is the spacetime counterpart of uniform circular motion — the curve of constant "curvature", generated by a one-parameter symmetry. It is also why Chapter 2.2's crew feel a constant g forever: their situation is literally unchanged by the passage of proper time, up to a boost.

(b) Asymptotes. As $|t| \rightarrow \infty$, $x = \sqrt{b^2 + c^2 t^2} \rightarrow c|t|$, so the asymptotes are $x = \pm ct$: **the light cone of the origin**. The rocket approaches the speed of light without reaching it, forever, hugging the cone from outside.

(c) Which flashes arrive. A flash emitted at $(t_0, 0)$ travels along $x = c(t - t_0)$. It reaches the rocket when

$$c(t - t_0) = \sqrt{b^2 + c^2 t^2}.$$

Square both sides — permissible provided $t > t_0$, which we then check:

$$c^2 t^2 - 2c^2 t t_0 + c^2 t_0^2 = b^2 + c^2 t^2 \implies t = \frac{c^2 t_0^2 - b^2}{2c^2 t_0}.$$

Note $c^2 t^2$ cancels: the equation is *linear* in t , so there is exactly one candidate meeting time.

Case $t_0 > 0$. Require $t > t_0$:

$$\frac{c^2 t_0^2 - b^2}{2c^2 t_0} > t_0 \iff c^2 t_0^2 - b^2 > 2c^2 t_0^2 \iff -b^2 > c^2 t_0^2,$$

which is impossible. *Case* $t_0 = 0$. The formula has zero in the denominator: no finite solution. *Case* $t_0 < 0$. Dividing by $2c^2 t_0$ now reverses the inequality, and the condition becomes $-b^2 < c^2 t_0^2$, which is always true. So the flash arrives — after a delay that grows without bound as $t_0 \rightarrow 0^-$.

Numbers, with $a = g$ and Chapter 2.2's $b = c^2/g = 0.9687$ ly: a flash at $t_0 = -1$ yr is caught at $t = (1 - 0.9384)/(-2) = -0.031$ yr; one at $t_0 = -0.1$ yr is caught at $t = +4.64$ yr; one at $t_0 = -0.01$ yr at $t = +46.9$ yr. The last light to make it takes forever to arrive, arbitrarily redshifted.

(d) The horizon. The dividing surface is the null line $x = ct$ — the future light cone of the origin, and simultaneously the asymptote from (b). Generalising the calculation to a flash from any event (t_e, x_e) : the ray $x = x_e + c(t - t_e)$ eventually overtakes the asymptote $x = ct$ if and only if $x_e - ct_e > 0$. So

the rocket can receive a signal from event $E \iff x_E > ct_E$.

Events on the far side of $x = ct$ are permanently invisible to it. That surface is the **Rindler horizon**.

What it is a horizon for, stated carefully. It is not a property of spacetime — this is flat Minkowski space, with no matter, no curvature, and no singularity anywhere. It is a property of the *observer*: it is the boundary of the region from which that particular eternally accelerating worldline can ever receive news. An inertial observer sails across it without noticing. Different accelerations a give different horizons; stop accelerating and yours vanishes, and all the delayed signals arrive.

■ Three forward pointers, all quoted. (i) The region $x > c|t|$ that the rocket can explore is called the *Rindler wedge*, and the coordinates adapted to the family of such observers are Rindler coordinates. (ii) Chapter 3.7 finds a horizon at the Schwarzschild radius of a black hole with exactly the same local character — a one-way surface with no local marker on it — and the argument that it is not a singularity is essentially the one above. (iii) Chapter 3.1 uses the equivalence principle to argue that a uniformly accelerated observer is locally indistinguishable from one at rest in a gravitational field, at which point the fact that acceleration alone can manufacture a horizon in *empty flat space* stops being a curiosity and becomes a clue.

PROBLEM 4 — COUNTING THE LORENTZ GROUP

(a) Verify that a spatial rotation $\Lambda = \text{diag}(1, R)$ with $R^T R = 1$ satisfies (2.3.13), so rotations are Lorentz transformations. (b) By counting independent equations in $\Lambda^T \eta \Lambda = \eta$, show that the solutions form a six-parameter family. (c) Identify the six as three boosts and three rotations by writing $\Lambda = 1 + \omega$ for infinitesimal ω and showing that $\omega_{\mu\nu} \equiv \eta_{\mu\rho} \omega^\rho{}_\nu$ is *antisymmetric*. (d) Connect the count to Chapter 2.4's arithmetic for antisymmetric tensors.

– Solution

(a) Rotations qualify. In block form with $\eta = \text{diag}(1, -1_3)$,

$$\Lambda^T \eta \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & R^T \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -R^T R \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \eta \quad \checkmark$$

Which is obvious in hindsight: a rotation leaves t alone and preserves $x^2 + y^2 + z^2$, so it preserves $c^2 t^2 - (x^2 + y^2 + z^2)$. The Lorentz group contains the rotation group as a subgroup — the transformations relating two observers who are at rest with respect to each other but have turned their axes.

(b) Six parameters. Λ is a 4×4 matrix: 16 unknown entries. The condition $\Lambda^T \eta \Lambda = \eta$ is an equation between two 4×4 matrices, so naively 16 equations — but both sides are *symmetric*. The left side is symmetric because $(\Lambda^T \eta \Lambda)^T = \Lambda^T \eta^T \Lambda = \Lambda^T \eta \Lambda$, using $\eta^T = \eta$; the right side is η . A symmetric 4×4 matrix equation carries only

$$\left. \frac{n(n+1)}{2} \right|_{n=4} = \frac{4 \times 5}{2} = 10$$

independent equations. Hence

$$16 - 10 = 6 \text{ free parameters.}$$

(The counting assumes the ten constraints are independent, which they are; (c) confirms it by exhibiting a six-dimensional solution space explicitly at the linear level.)

(c) Three and three. Put $\Lambda = 1 + \omega$ with ω small and keep first order:

$$(1 + \omega^T) \eta (1 + \omega) = \eta + \omega^T \eta + \eta \omega + O(\omega^2) = \eta \implies \omega^T \eta + \eta \omega = 0.$$

Define the matrix $\hat{\omega} \equiv \eta \omega$, whose entries are $\omega_{\mu\nu} = \eta_{\mu\rho} \omega^\rho{}_\nu$ — the index has been lowered, in the sense Chapter 2.4 §4.1 makes precise. Then

$$\hat{\omega}^T = \omega^T \eta^T = \omega^T \eta = -\eta \omega = -\hat{\omega}.$$

So $\omega_{\mu\nu} = -\omega_{\nu\mu}$: **antisymmetric**. Its independent entries are the strictly-upper-triangular ones, and they split naturally:

- ω_{0i} for $i = 1, 2, 3$ — mixing time with a space direction. **Three boosts**, one per axis.
- ω_{ij} for $1 \leq i < j \leq 3$ — mixing two space directions. **Three rotations**, one per plane xy, yz, zx .

Total $3 + 3 = 6$, matching (b) ✓. Check the shape against Chapter 2.2's boost: expanding $\Lambda(\phi)$ to first order in ϕ gives $\omega^0_1 = \omega^1_0 = -\phi$, and lowering the first index with η turns the second of these into $\omega_{10} = -\eta_{11}\phi = +\phi$ while $\omega_{01} = -\phi$ — antisymmetric ✓. Note that ω^μ_ν itself is *not* antisymmetric; it is only after lowering that the property appears, which is a first hint that index position carries meaning.

(d) The connection to Chapter 2.4. Chapter 2.4 §7.2 counts the independent components of an antisymmetric rank-2 tensor in n dimensions as $\frac{1}{2}n(n-1)$, giving $\frac{1}{2} \cdot 4 \cdot 3 = 6$ in spacetime — and remarks that six is not four, so such an object cannot masquerade as a four-vector the way **B** masquerades in three dimensions. Here is the same six, arrived at completely independently: not by counting slots in a tensor, but by counting how many ways there are to move while preserving η .

They are the same six for a reason. The generators of the Lorentz group *are* an antisymmetric rank-2 object, and Chapter 2.6 will find that the electromagnetic field tensor $F^{\mu\nu}$ is another one with the same six slots — three of **E**, three of **B**. ▮ That the match is structural rather than coincidental is Chapter 6.1's business: the field strength of a gauge theory lives in the Lie algebra of its symmetry group, and for Lorentz symmetry that algebra is exactly the antisymmetric matrices counted above.

THE BRICK YOU JUST LAID

You turned Chapter 2.2's algebra into a geometry. The interval $\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$ is invariant, proved by direct substitution; the Lorentz transformations are *defined* as the linear maps preserving it, exactly as rotations are the maps preserving $x^2 + y^2 + z^2$; and the whole subject is Euclidean geometry with one sign changed. You met x^μ , $\eta_{\mu\nu}$, the summation convention, and $\Lambda^\top \eta \Lambda = \eta$ — informally, with the debt to Chapter 2.4 recorded. You saw that boosts are hyperbolic rotations, that the rapidity is the hyperbolic angle and adds because areas add, that the orbits are the invariant hyperbolae, and that those hyperbolae calibrate every observer's axes and explain why the page misleads. You classified separations as timelike, null or spacelike, proved the classification and the causal order are invariant, proved that spacelike order is always reversible, and built a closed causal loop from two superluminal signals — which is why c is a limit on causation and not a fact about light. You defined proper time as the length of a worldline, proved that the straight worldline *maximises* it, both directly and variationally, and identified the twin paradox as the reverse triangle inequality. And you met the four-velocity, an object of permanently fixed length c .

Where this gets spent. Chapter 2.4 takes every piece of notation introduced here on trust — x^μ , $\eta_{\mu\nu}$, upper and lower indices, "transforms like dx^μ " — and turns it into definitions, with the transformation law as the definition of a tensor; its opening sentence is about the debt this chapter incurred. Chapter 2.5 feeds the four-velocity of §7 into $p^\mu = mu^\mu$, gets $p \cdot p = m^2 c^2$ from (2.3.47) in one line, and reads $E = mc^2$ off it. Chapter 2.6 finds the six components counted in Problem 4 sitting inside the electromagnetic field tensor. Chapter 3.1 keeps the light cone and throws away everything else, which is what "spacetime is locally Minkowski" means. Chapter 3.2 builds the manifold on which that statement can be made. **Chapter 3.3 takes (2.3.36) unchanged, replaces $\eta_{\mu\nu}$ by $g_{\mu\nu}(x)$, and calls the extremal paths gravity** — the single most important thing this chapter sets up. Chapter 3.7 meets Problem 3's horizon again, this time around a black hole. And Chapter 7.2 returns to the observation that the invariance group of a metric is where the physics is, in two dimensions, where that group turns out to be infinite-dimensional and the string becomes solvable because of it.